Set 2 : Tests 11 to 20

11. AVM Bandra, Mumbai
12. Hiranandani Foundation, Thane
13. Nanavati, Mumbai
14. NES National Public, Mumbai
15. Vibgyor High, Mumbai
16. Vissanji, Mumbai
17. N L Dalmia, Mumbai
18. Gundecha, Mumbai
19. Universal, Mumbai
20. Children’s Academy, Mumbai

Question Paper 11

PRE- PRELIMINARY REVIEW -

Std: X **Marks : 80**

MATHEMATICS **Time : 3 hours**

Answers to this Paper must be written on the paper provided separately.
You will not be allowed to write during the first 15 minutes.
This time is to be spent in reading the question paper.
The time given at the head of this Paper is the time allowed for writing the answers.

Attempt all questions from Section A and any four from Section B
All working, including rough work, must be clearly shown and must be done on the same sheet as the rest of the answer.
Omission of essential working will result in the loss of marks.
The intended marks for questions or parts of questions are given in brackets [ ].
This paper consists of **thirteen** printed pages.

SECTION - A (40 MARKS)

(Attempt all questions from this section)

Question 1:

Choose the correct answer to the following questions from the given options : [15]

(i) A manufacturer sells goods to a wholesaler for ₹20,000 with GST @ @12%. The wholesaler sells to a retailer for ₹24,000 with GST @12%. Find the net tax payable by the wholesaler.

(a) Rs. 240

(b) Rs. 480

(c) Rs. 288

(d) Rs. 538

..2

(ii) The value of p for equation $2x^2 - 4x + p = 0$ to have real roots will be _____.

(a) $p \leq -2$

(b) $p \geq 2$

(c) $p \leq 2$

(d) $p \geq -2$

(iii) The coordinates of the centroid of a triangle whose vertices are (0, 6), (8,12) and (8, 0) is _____.

(a) (4, 6)

(b) (16/3, 5)

(c) (8, 6)

(d) (16/3 , 6)

(iv) If a, b, c are in continued proportion, find the correct relation:

(a) $b^2 = ac$

(b) $a^2 = bc$

(c) $c^2 = ab$

(d) $ab = ac$

(v) If a point P is equidistant from two intersecting lines, L1 and L2, its locus is:

(a) The perpendicular bisector of the segment connecting L1 and L2.

(b) The line parallel to L1.

(c) The angle bisector of the angle formed by L1 and L2.

(d) A circle centered at the intersection of L1 and L2.

(vi) The value of : $\tan^2 A - \left(\frac{1}{\cos A} X \sec A\right)$

(a) 1

(b) 0

(c) -1

(d) 2

(vii) The circumference of the base of a right circular cylinder is 44cm. If the height of the cylinder is 10cm, what is its Curved Surface Area .

(a) 220 cm^2

(b) 440 cm^2

(c) 880 cm^2

(d) 1540 cm^2

(viii) **Assertion (A):** For an odd number of observations, the median is the value of the $\frac{n+1}{2}th$ term when data is arranged.

Reason (R): For the data 12, 4, 9, 15, 7 , the median is 9

(a) Both A and R are correct, and R is the correct explanation of A.

(b) Both A and R are correct, but R is not the correct explanation of A.

(c) A is true, but R is false

(d) Both A and R are false

(ix) If the probability of an event A is 0.65, then the probability of not A is:

(a) 0.35

(b) 0.65

(c) 0.55

(d) 0.45

(x) Assertion (A): The product of a matrix 2 x 3 and a matrix 3 x 2 is a square matrix.
Reason (R): The number of columns of the first matrix must equal the number of rows of the second matrix for multiplication to be defined.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

(xi) The mean proportion of:(a – b) and ($a^3 - a^2b$) , a > b

(a) a (b – a)

(b) a (a – b)

(c) -a (a – b)

(d) a (-a – b)

(xii) Two roads intersect such that one has the equation y = 2x +1. If a new road must be perpendicular to it and passes through the point (2, 3), what is its equation?

(a) $y - 3 = \frac{3}{2}(x - 2)$

(b) $y - 3 = \frac{-1}{2}(x - 2)$

(c) $y - 3 = -2(x - 2)$

(d) $y = 12x + 3$

(xiii) A point P (-3 , 4) is reflected in the line x = 2, the coordinates of the point of reflection P' are _____.

(a) (7,3)

(b) (5, 3)

(c) (7, 4)

(d) (3, 4)

(xiv) What does the symbol n(n+1) represent in the numerator of the interest formula for a recurring deposit?

(a) The total principal deposited over n months

(b) The sum of the time periods for which interest is calculated on each installment.

(c) The maturity amount after n months.

(d) The total rate of interest multiplied by the time period.

(xv) A carton manufacturing company uses the volume of its standard box, $V(x) = x^3 + 5x^2 - 2x - 24$, to represent its dimensions. Which of the following expressions must be a linear dimension (factor) of the box?

(a) $(x + 1)$

(b) $(x-1)$

(c) $(x + 3)$

(d) $(x + 2)$

Question 2

(i) Usha invests ₹6,000 in 8% ₹10 shares available at ₹20. Later, she sells all the shares when the market price rises to ₹32. She then invests the entire sale proceeds in 12% ₹100 shares available at ₹120 each. Calculate: [4]

(i) Her sale proceeds

(ii) The number of ₹120 shares she buys

(iii) The change in her annual dividend income.

(ii) If $(x - 3)$ is a factor of the polynomial $P(x) = x^3 - 3x^2 - px + 3$. Find the value of p and factorize the expression completely. [4]

(iii) The marks scored by 750 students in an examination are given in the form of a frequency distribution table: (on x-axis 2 units=40 marks, Y-axis = 100 students) [4]

Marks	No. of Students
600 – 640	15
640 – 680	45
680 – 720	155
720 – 760	285
760 – 800	170
800 – 840	65
840 – 880	15

Draw an Ogive and answer the following question:-

a) The Median Marks

b) The number of students who scored more than 700 marks in the examination.

Question Paper 12

First Preliminary Examination–

Subject: Mathematics

Std: X **Time: 3 hrs**

Max.Marks:80

Answers to this Paper must be written on the paper provided separately. You will not be allowed to write during first 10 minutes. This time is to be spent in reading the question paper. The time given at the head of this Paper is the time allowed for writing the answers. Attempt ***all*** *questions from* ***Section A*** *and* ***any four*** *questions from* ***Section B****. All working including rough work, must be clearly shown, and must be done on the same sheet as the rest of the answer.*

Omission of essential working will result in loss of marks. The intended marks for questions or parts of questions are given in [ ]. This Question paper consists of ***6*** *printed pages.*

Mathematical tables are provided.

SECTION A

(Attempt all questions from this section)

Question 1 [15]

Choose the correct answer to the questions from the given options.

(Do not copy the questions, write only the answers)

(i) The probability of getting a number divisible by 3 in throwing a die is

a) $\frac{1}{6}$ b) $\frac{1}{3}$ c) $\frac{1}{2}$ d) $\frac{2}{3}$

(ii) If $3y - 2x - 4 = 0$ and $4y - ax - 2 = 0$ are perpendicular to each other, then the value of a is

a) 6 b) 7 c) -7 d) -6

(iii) The following numbers 1, 1, 3, x, 6, 6, 7 are arranged in an ascending order. If their mean and median are same, then the value of x is

a) 4 b) 3 c) 5 d) 6

(iv) The surface area of a sphere is 154 cm^2. The volume of the sphere is

a) $359\frac{1}{3}cm^3$ b) $736\frac{1}{3}cm^3$ c) $179\frac{2}{3}cm^3$ d) $1437\frac{1}{3}cm^3$

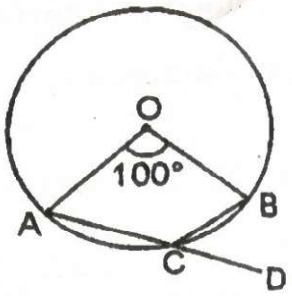
(v) If the point P (a, b) is reflected in the origin and then reflected in the x axis to the point P" (12, -5), then the value of (b – a) is

a) -17 b) 17 c) 7 d) -7

(vi) If $f(x) = 2x^3 + ax^2 + 3x - 5$ leaves remainder 33 when divided by $x - 2$, then the value of a is

a) 5 b) 3 c) 2 d) 4

(vii) **Assertion** (A); In the given figure, if O is the centre of the circle, then $\angle BCD = 80°$.
Reason (R) ; Exterior angle of a cyclic quadrilateral is equal to the interior opposite angle.



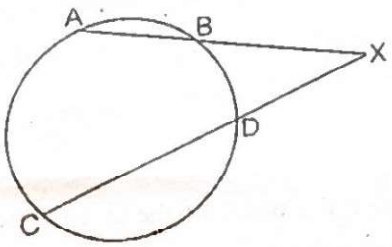
a) (A) is true and (R) is false

b) (A) is false and (R) is true

c) Both (A) and (R)are true and (R) is the correct explanation of (A)

d) Both (A) and (R)are true and (R) is not the correct explanation of (A)

(viii) In the given figure, if AB = 4 cm, BX = 6 cm and XD = 5 cm , then the length of CD is



a) 5 b) 6 c) 7 d) 3

(ix) The total bill of a car service given by Mr. Vikas is ₹6720 including tax. If the rate of GST is 12%, then the actual amount of the bill excluding tax is

a) ₹6000 b) ₹5800 c) ₹7000 d) ₹7526.40

(x) The roots of the quadratic equation $9x^2 - 5x + 1 = 0$ are

a) two equal real roots b) two distinct real roots
c) more than two real roots d) no real roots

(xi) **Assertion** (A): $\Delta ABC \sim \Delta DEF$, such that area($\Delta ABC$)= 36 cm^2 and area(ΔDEF) = 49 cm^2, then AB: DE = 6 : 7

Reason (R) : If $\Delta ABC \sim \Delta DEF$, then $\frac{\text{area}(\Delta ABC)}{\text{area}(\Delta DEF)} = \frac{AB^2}{DE^2} = \frac{BC^2}{EF^2} = \frac{AC^2}{DF^2}$

a) (A) is true and (R) is false

b) (A) is false and (R) is true

c) Both (A) and (R)are true and (R) is the correct explanation of (A)

d) Both (A) and (R)are true and (R) is not the correct explanation of (A)

(xii) If $cos\theta = \frac{2}{3}$, then the value of $2\,sec^2\theta + 2\,tan^2\theta - 7$ is

(xiii) If $\frac{8x+13y}{8x-13y} = \frac{9}{7}$, then the value of $x : y$ is

a) 13 : 1 b) 26 : 1 c) 1 : 13 d) 1 : 5

(xiv) Which term of the G.P.$\sqrt{3}$, $3\sqrt{3}$, $9\sqrt{3}$,.....is $729\sqrt{3}$?

a) 7th b) 6th c) 9th d) 8th

(xv) If M x $\begin{bmatrix} 3 & -4 \\ 0 & 7 \end{bmatrix}$ = [6 17], then the order of the matrix M is

a) 1 x 2 b) 2 x 2 c) 1 x 1 d) 2 x 1

Question 2

(i) If $x+1$ and $x-2$ are the factors of $x^3 + ax^2 - bx - 6$, then find the values of a and b. With these values of a and b, factorise the given polynomial completely. [4]

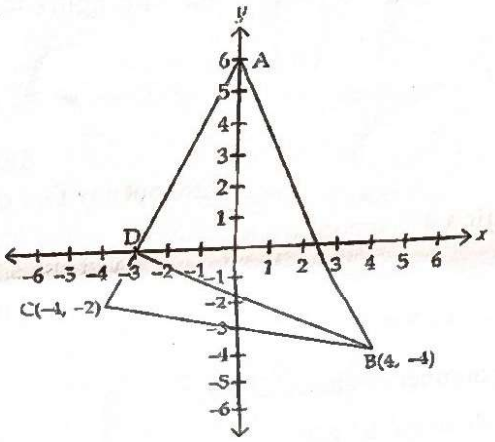
(ii) Observe the given figure and find [4]

a) the coordinates of A and D

b) the coordinates of the centroid of ΔABC

c) the ratio in which D divides the line segment AC

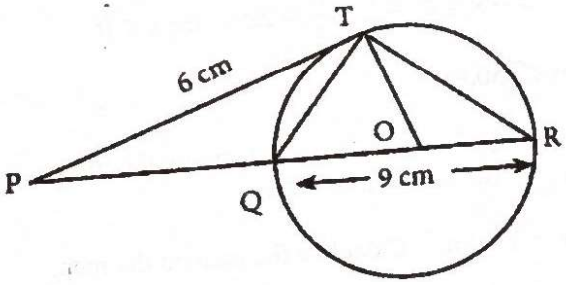
d) the equation of line BD



(iii) Observe the given figure and answer the following. [4]

a) Prove that $\Delta PQT \sim \Delta PTR$

b) If PT = 6 cm, QR=9 cm, then find the length of PQ.



Question 3

i. The ninth term of an A.P.is equal to seven times the second term and twelfth term exceeds five times the third term by 2. Find [4]

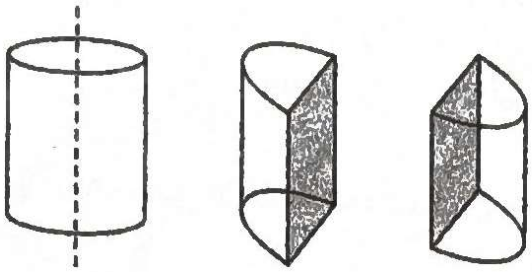
a) the first term and the common difference

b) the sum of first 18 terms

ii. A solid metallic cylinder is cut into two identical halves along its height (as shown in the diagram). The diameter of the cylinder is 7 cm and the height is 10 cm. Find [4]

a) the total surface area (both the halves)

b) the total cost of painting the two halves at the rate of ₹ 30 per cm^2



(iv)Use a graph paper for this question

[5]

a) Plot the points A(0, -4), B(3, -2), C(1, 0), D(3, 2) and E(0, 1)

b) Reflect the points B, C and D on the y axis and name them as B', C' and D' respectively.

c) Write the coordinates of B', C' and D'

d) Write the geometrical name the figure formed by joining EDCBA B'C' D'E

SECTION B

(Attempt any four questions from this section)

Question 4

i. A man invests ₹50,000 in ₹100 shares available at 50% discount. If the company declares 5% dividend, find [3]

a) the number of shares he buys

b) the dividend he gets

c) the rate of return on his investment

ii. Solve the following inequation, write the solution set, and represent it on the real number line. [3]

$$-3 + x \leq \frac{8x}{3} + 2 \leq \frac{14}{3} + 2x, \text{ where } x \in R$$

iii. A map is drawn to a scale of 1 : 50,000. [4]

a) An airport runway is represented by a line of length 4.6 cm on the map. Calculate the actual length of the runway in kilometers.

b) The actual area of the airport is 3.5 km^2. Calculate the area on the map.

Question 5

i. The hotel bill for a number of people for overnight stay is ₹4,800. If there were four more people, the bill each person had to pay would have reduced by ₹200. Find the number of people staying overnight. [3]

ii. Mrs. Sen deposits a sum of ₹ 2,400 every month in a recurring deposit scheme of a bank. If she receives ₹ 30,048 at 8% p.a. interest rate , calculate the term (in years) for which the account was held. [3]

iii. Prove the following trigonometry identity: [4]

$$\frac{tan^3\theta}{1+tan^2\theta}+\frac{cot^3\theta}{1+cot^2\theta}=sec\theta cosec\theta-2sin\theta cos\theta$$

Question 6

i. Using step-deviation method, find mean for the following frequency distribution. Write your answer to the nearest whole number. [3]

Weight in grams	70 - 75	75 - 80	80 - 85	85 - 90	90 - 95	95 - 100	100 - 105	105 - 110
No. Of Apples	5	8	10	9	3	8	4	3

ii. Two circles of radii 50 cm and 18 cm touch each other externally. Find the length of the direct common tangent. [3]

iii. Mohan buys a washing machine from a wholesaler for ₹30,000. He marks the price of the washing machine 10% above the cost price and sells it to Soham at a discount of 5% on the marked price. If the sale is intrastate and the rate of GST is 12%, find [4]

a) the marked price of the washing machine

b) the amount which Soham pays to Mohan for the washing machine including GST

c) the amount of tax payable by Mohan to the Central Government.

Question 7

i. The angle of elevation of an aeroplane from a point on the ground is 50°. After 6 seconds, the angle of elevation changes to 60°. If the plane is flying horizontally at a height of 3000 m, find the speed of the plane in km/h. [5]

ii. The scores of 120 shooters in a shooting competition were recorded as follows: [5]

Scores obtained	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50	50 - 60	60 - 70	70 - 80	80 - 90	90-100
No. of shooters	5	9	16	22	26	18	11	6	4	3

Using a graph sheet draw ogive for the given data and use it to find

a) the median

b) the interquartile range

c) the number of shooters who obtained more than 75% scores

Question 8

i. If the roots of the equation $(a^2 + b^2)x^2 - 2(ac + bd)x + (c^2 + d^2) = 0$ are equal, then prove that ad = bc. [3]

ii. A bag contains 15 balls out of which x are blue and the remaining are red. [3]

a) If a ball is drawn at random, what is the probability of getting a red ball (in terms of x)

b) If 5 more red balls are put in the bag and a ball is drawn at random, then the probability of getting red ball is double the probability obtained in a). Find x.

iii. If $x = \frac{\sqrt{a^2+b^2}+\sqrt{a^2-b^2}}{\sqrt{a^2+b^2}-\sqrt{a^2-b^2}}$, by using properties of proportion, prove that $b^2 = \frac{2a^2x}{x^2+1}$ [4]

Question 9

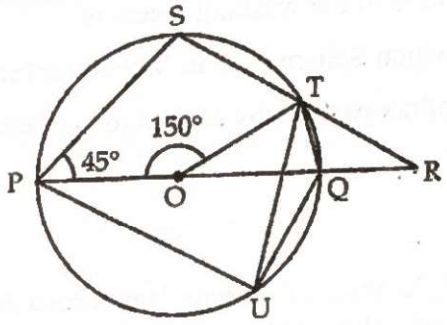
i. Use a graph sheet for this question. Draw a histogram for the following data and estimate the mode from it. [3]

Marks	10 - 19	20 - 29	30 - 39	40 - 49	50 - 59
No. of students	6	9	16	13	4

ii. Fine the number of the terms in the G.P. 3, $\frac{3}{2}$, $\frac{3}{4}$,.....if the sum of the given terms is $\frac{186}{32}$. [3]

iii. Observe the given figure where O is the center of the circle and find the value of [4]

a) ∠PST
b) ∠PUT
c) ∠QTR
d) ∠QRT.



Question 10

i. If b is the mean proportion between a and c, prove that $\frac{a^4+a^2b^2+b^4}{b^4+b^2c^2+c^4} = \frac{a^2}{c^2}$ [3]

ii. If $f(x) = x^2 - 5x + 7I$, where I is an identity matrix and A= $\begin{bmatrix} 1 & 0 \\ -1 & 7 \end{bmatrix}$, then find $f(A)$. [3]

iii. Solve $x^2 + 7x = 7$ and give your answer correct to three significant figures. [4]

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Question Paper 13

FIRST PRELIMINARY EXAMINATION
MATHEMATICS

Grade: 10
Marks: 80

Date:
Reading Time: 15 minutes
Writing Time: 3 hours

Instructions:

*Attempt **all** questions from **Section A** and **any four** questions from **Section B**.*
All working, including rough work, must be clearly shown and must be done on the same sheet as the rest of the answer.

Omission of essential working will result in loss of marks.

The intended marks for questions or parts of questions are given in brackets [ ]

Mathematical tables are provided.

SECTION A (40 marks)

*(Attempt **all** questions from this Section)*

Question 1

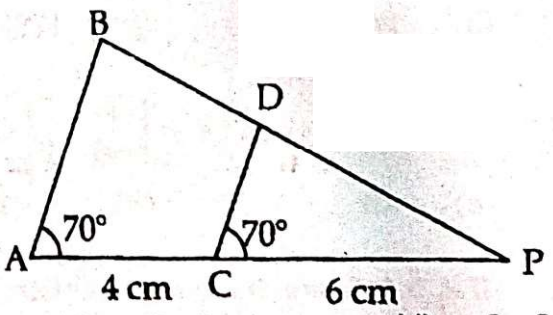
Choose the correct answers to the questions from the given options. **[15]**

(Do not copy the question, write the correct answers only.)

(i) $(2x + 1)$ and $(x - 3)$ are the factors of $2x^3 - 9x^2 + 7x + 6$. The third factor of the given polynomial is:

(a) $(x + 2)$ (b) $(x - 1)$ (c) $(x - 2)$ (d) $(x + 1)$

(ii) In the given figure, PD : PB is:



(a) 3 : 5 (b) 3 : 2 (c) 5 : 3 (d) 2 : 3

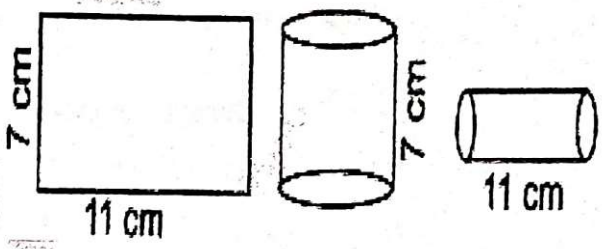
(iii) Giridhar deposited ₹600 per month for 30 months in a recurring deposit account at a certain rate. If he earns ₹2,500 as interest at the time of maturity, then his matured amount is:

(a) ₹ (600 × 30 × 31 + 2,500)
(b) ₹ (600 × 15 × 31 + 2,500)
(c) ₹ (600 × 30 + 2,500)
(d) ₹ (600 × 31 + 2,500)

(iv) At one end 'A' of a diameter AB of a circle of radius 5 cm, tangent XAY is drawn to the circle. The length of the chord CD parallel to XY and at a distance 8 cm from 'A' is:

(a) 8 cm (b) 6 cm (c) 5 cm (d) 4 cm

(v) A rectangular sheet of paper of size 11 cm × 7 cm is first rotated about the side 11 cm and then about the side 7 cm to form a cylinder, as shown in the figure.



Assertion (A): The ratio of their curved surface areas is 2 : 1.

Reason (R): Curved surface area of a cylinder = $2\pi rh$

(a) A is true, R is false.
(b) R is true, A is false.
(c) Both A and R are true, and R is the correct reason for A.
(d) Both A and R are true, and R is the incorrect reason for A.

(vi) For the data 5, 9, 7, 10, 13, 11, 10, the median is:

(a) 10 (b) 11 (c) 13 (d) 7

(vii) The probability of no voter turnout in a General Election in India is P(S).
The value of P(S) =

(a) P(S) < 0 (b) P(S) = 0 (c) P(S) = 1 (d) P(S) > 1

(viii) If $A = [0 \quad 3]$ and $B = \begin{bmatrix} 1 & 2 \\ -6 & 2 \end{bmatrix}$ which of the following operation is possible?

(a) AB (b) A − B (c) A + B (d) BA

(ix) If x, 5.4, 5, and 9 are in proportion then x is:

(a) 3 (b) $\frac{25}{3}$ (c) 9.72 (d) 25

(x) **Assertion (A):** $\tan^2\theta + \cos^2\theta - 1 = \tan^2\theta . \sin^2\theta$

Reason (R): $1 - \cos^2\theta = \sin^2\theta$

(a) Both A and R are true, and R is the incorrect reason for A

(b) Both A and R are true, and R is the correct reason for A.

(c) A is true, R is false.

(d) R is true, A is false.

(xi) Nakul bought an electrical fan which has a marked price of ₹2,000. If the GST on the goods is 5%, then the SGST will be:

(a) ₹40 (b) ₹80 (c) ₹50 (d) ₹100

(xii) A man invested in a company paying 8% dividend on its share. If the % return on his investment is 16%, then the shares are

(a) Cannot be determined (b) At par (c) At premium (d) At discount

(xiii) One of the roots of the quadratic equation $x^2 - 8x + 5 = 0$ is 7.3166. The root of the equation correct to 4 significant figures is:

(a) 7.3166 (b) 7.32 (c) 7.316 (d) 7.317

(xiv) Which of the following equation represents a line with a negative slope?

(a) $x - y = 7$ (b) $2x + 3y + 7 = 0$ (c) $y = -7$ (d) $x = 7$

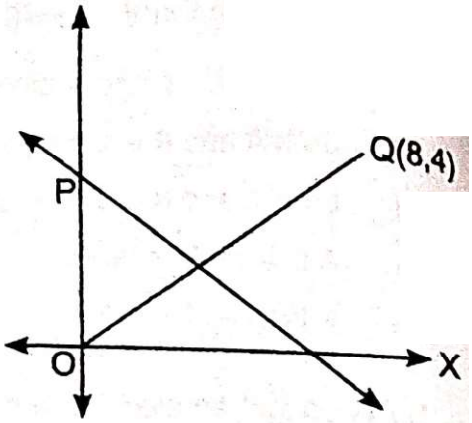
(xv) If $\frac{1}{x-2} < 1$, and x is a composite number. The solution set for 'x' is:

(a) {1, 2} (b) {2} (c) {0} (d) Ø

Question 2

(i) Find the value of 'k' if $(x + 2)$ is a factor of $3x^3 + kx^2 - 18x + 40 = 0$. Hence, factorize the expression. **[4]**

(ii) *Based* on the given figure find:

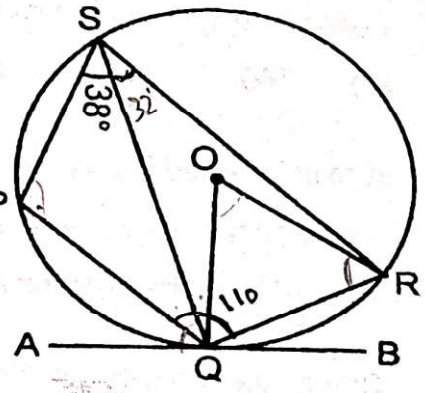


(a) The coordinates of point P where line $2x + y = 6$ meets y-axis.

(b) The equation of line OQ where O is the origin and Q = (8, 4).

(c) State with reason if the two lines are perpendicular to each other.

(iii) AB is a tangent to the circle at Q. O is the centre of the circle. PQRS is a cyclic quadrilateral. If ∠PSQ = 38°, ∠PQR = 110°, find: **[4]**



(a) ∠QPR.

(b) ∠PQA.

(c) ∠QRO.

Question 3

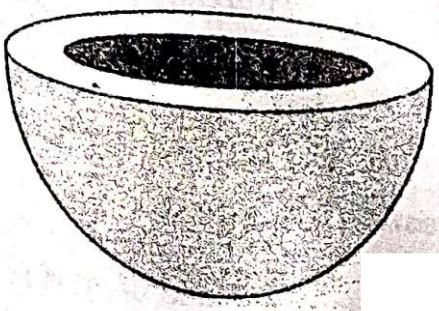
(i) $\frac{-5}{3}, \frac{-4}{3}, -1, \frac{-2}{3}, \ldots\ldots 4\frac{2}{3}$ are the terms of a progression. **[4]**

(a) What is the type of progression? Justify.

(b) How many terms are there in all?

(c) Find the two middle most terms of the progression.

(ii) A spherical watermelon of radius 4 cm is halved. The uniform width of the white part surrounding the red edible part is 1 cm. From half of the melon, the red part of the fruit is scooped out. Taking $\pi = 3.14$, find: **[4]**



(a) The volume of the edible part of the fruit taken out (ignore the volume of the seeds).

(b) The surface area of the white part of that hemispherical piece.

(iii) Use a ruler and compass to answer this question. **[5]**

(a) Construct a circle of radius 4.5 cm and draw a chord AB of length 6.5 cm.

(b) At A, construct $\angle CAB = 75°$, where C lies on the circumference of the circle.

(c) Construct the locus of all points equidistant from A and B.

(d) Construct the locus of all points equidistant from CA and BA.

(e) Mark the point of intersection of the two loci as P. Measure and write down the length of CP.

SECTION B (40 marks)

*(Attempt **any four** questions from this section)*

Question 4

(i) Ruchi invested ₹8,000 in 7% ₹100 shares at ₹80. After a year she sold these shares at ₹75 each and invested the proceeds (including her dividend) in 18%, ₹25 shares at ₹41. Find: **[3]**

(a) Dividend for the first year.

(b) % increase in return on her original investment.

(ii) Solve the following inequation, write the solution set, and represent it on the real number line: **[3]**

$$2 - x \leq 5 - 2x < x - 1, x \in R$$

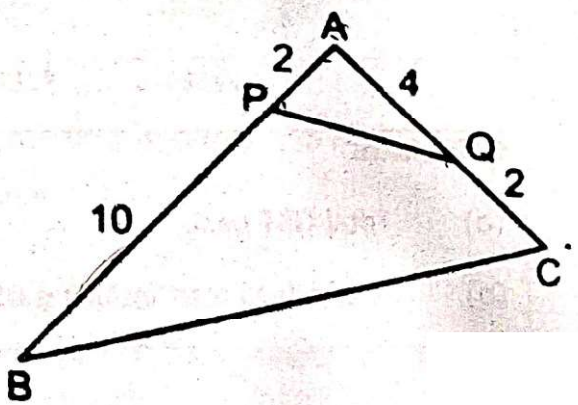
(iii) Prove that: $\frac{\cot A + cosec\ A - 1}{\cot A - cosec\ A + 1} = \frac{1 + \cos A}{sin\ A}$ **[4]**

Question 5

(i) In Δ ABC, P and Q are points on AB and AC such that AP = 2 cm, PB = 10 cm, AQ = 4 cm and QC = 2 cm. **[3]**

(a) Prove that Δ APQ ~ Δ ACB.

(b) Find area of Δ APQ, if area of Δ ABC = 27 cm².



(ii) Neha opens a recurring deposit account of ₹600 per month at 12% p.a. if she is paid ₹7,668 as maturity amount, how many installments does she need to pay? **[3]**

(iii) Calculate the mean of the following distribution by step deviation method. **[4]**

Class	10–19	20–29	30–39	40–49	50–59
Frequency	8	10	20	7	5

Question 6

(i) Point A divides the join of line segment M (3, –2) and N (10, 19) in the ratio 3 : 4. **[3]**

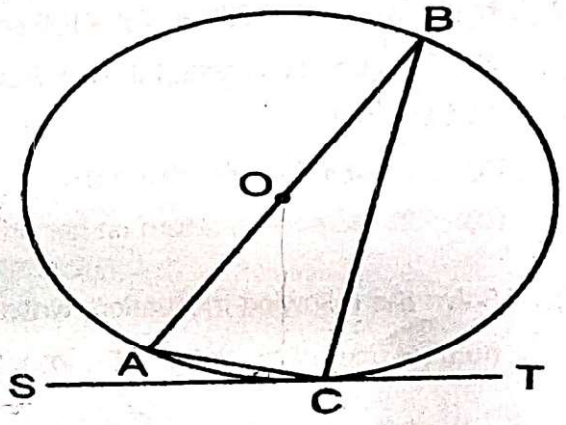
(a) Find the coordinates of point A.

(b) The equation of a line passing through A and parallel to $2x + 3y = 1$

(ii) SCT is a tangent to the circle with centre O. AB is a diameter, if ∠ACS = 15°, find: **[3]**

(a) ∠ABC.

(b) ∠CAB.



(iii) The following bill shows the GST rate and the marked price of articles: **[4]**

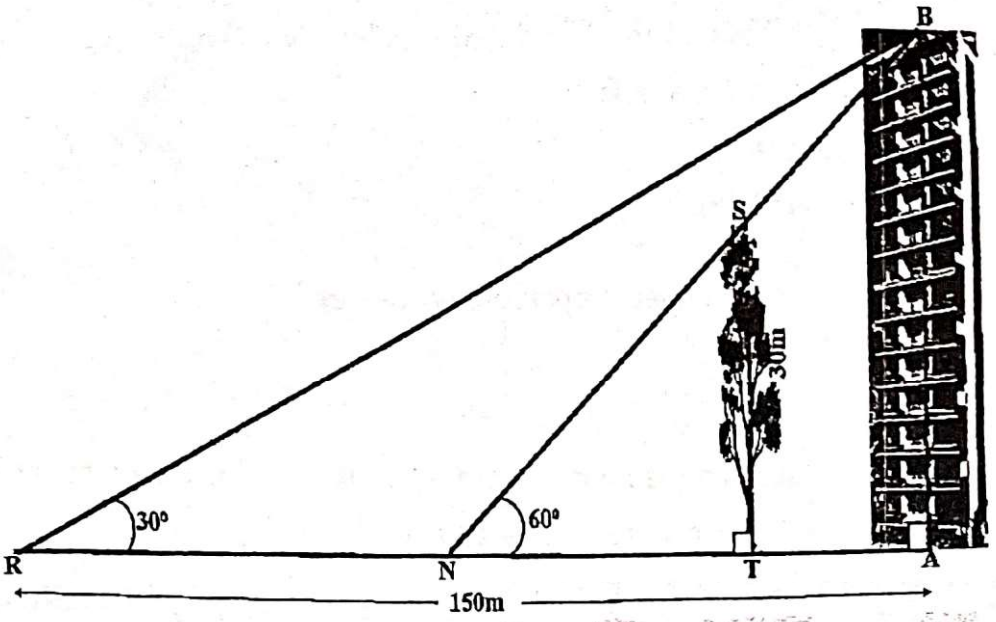
Indraprastha Sweet Mart				
S.No.	**Item**	**Price**	**Quantity**	**Rate of GST**
1.	Laddu	₹500 per kg	2 kg	5%
2.	Pastries	₹100 per piece	12 pieces	18%

Calculate:

(a) Total GST paid.

(b) Total bill amount including GST.

Question 7



[5]

(i) A tree (TS) of height 30 m stands in front of a tall building (AB). Two friends Messi and Neymar are standing at R and N respectively, along the same straight line joining the tree and the building (as shown in the diagram). Messi, standing at a distance of 150 m from the foot of the building, observes the angle of elevation of the top of the building as 30°. Neymar from his position observes that the top of the building and the tree has the same elevation of 60°. Find the:

(a) Height of the building.

(b) Distance between Neymar and the foot of the building.

(c) Distance between Messi and Neymar.

(d) Distance between building and the tree.

(ii) The marks of an examination are tabulated below: **[5]**

Marks (less than)	10	20	30	40	50	60	70	80	90	100
No. of candidates	8	20	40	75	125	160	185	192	197	200

Take 2 cm = 10 marks on one axis and 2 cm = 20 students on the other, draw an ogive for the above data and from it determine:

(a) The median marks.

(b) The number of candidates who failed if pass mark is 35.

(c) The number of candidates who obtained grade A, if the lowest mark for Grade A is 75.

Question 8

(i) Cards are marked from 10 to 50 and well shuffled. One card is drawn at random. What is the probability that it is a: **[3]**

(a) A perfect square.

(b) A multiple of 3 and 4.

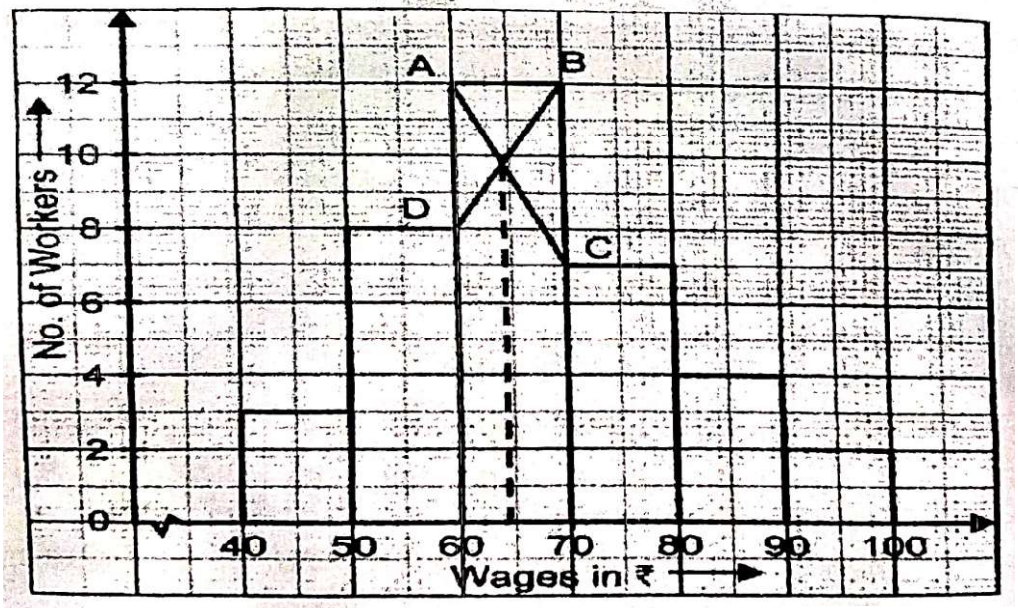
(ii) If a, b, and c are in continued proportion, prove that: **[3]**

$$\left(\frac{ab + bc + ac}{a + b + c}\right)^3 = abc$$

(iii) Some wheat is placed on the floor in a conical shape of diameter 4.8 m and height 1.5 m. This wheat is transferred into a cylindrical drum of diameter 2.4 m and is completely filled. Find the height of the drum. **[4]**

Question 9

(i) Study the graph given below and answer the following (scale 1 cm = ₹10 on x-axis and 1 cm = 2 workers on y-axis): **[3]**



(a) Number of workers earning ₹50 to ₹70.

(b) Modal class interval.

(c) The value of the mode.

(ii) How many terms of the G.P. 1, 3, 9, 27,.....will make the sum of 1093? **[3]**

(iii) If $x = 2$ is one root of the equation $(k - 3)x^2 - kx - 8 = 0$, find the value of 'k'. Also, find the other root of the equation. **[4]**

Question 10

(i) By increasing the speed of a car by 10 km/h, the time of journey for a distance of 72 km is reduced by 36 minutes. Find the original speed of the car. **[3]**

(ii) Given $A = \begin{bmatrix} p & 0 \\ 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 9 & 0 \\ 5 & 4 \end{bmatrix}$. If $A^2 = B$, find p. **[3]**

(iii) For this question, use a graph paper. Scale: 1 cm = 1 unit along both x and y-axis. Plot the points A (2, 2), and B (6, –2) in the graph and answer the following: **[4]**

(a) Reflect point A in origin to point D and write the co-ordinates of point D.

(b) Reflect point A in line $y = -2$ to point C and write the co-ordinates of point C.

(c) Find a point P on CD which is invariant under reflection in $x = 0$, write its coordinates.

(d) Write the geometrical name of the closed figure ABCD.

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Question Paper 14

First Preliminary Examination –

Class: X
Sub: Mathematics
Name: _

Marks: 80
Time: 3 hours

Instructions:

1. *Answers to this Paper must be written on a single line paper separately.*
2. *You will not be allowed to write during the first 15 minutes. This time is to be spent in reading the question paper.*
3. *The time given at the head of this Paper is the time allowed for writing the answers.*
4. *The intended marks for questions or parts thereof are given in brackets.*
5. *Section I is compulsory. Attempt any four questions from section II.*
6. *Rough work to be done in the rough column.*
7. *Draw a neat diagram with pencil wherever required.*

SECTION I (40 Marks)
Attempt all questions

Question 1: Choose the correct answer from the following: [15]

(i) If $x\sin^3\theta + y\cos^3\theta = \sin\theta\cos\theta$ and $x\sin\theta = y\cos\theta$, then $x^2 + y^2$ is equal to:

(a) 0 (b) ½ (c) 1 (d) 3/2

(ii) Which of the following points is invariant with respect to the line $y = -2$?

(a) (3,2) (b) (3,-2) (c) (2,3) (d) (-2,3)

(iii) Ruhee invests ₹9600 on ₹100 shares at ₹80 if the company pays him 18% dividend then his percentage return on shares is:

(a) 22.0% (b) 22.5% (c) 20.5% (d) 20.6%

(iv) The values of x satisfying $p^2x^2 + (p^2 - q^2)x - q^2 = 0$ are:

(a) $\frac{q^2}{p^2}, -1$ (b) $\frac{p^2}{q^2}, 1$ (c) $1, \frac{-q}{p}$ (d) $\frac{q^2}{p^2}, 1$

(v) The point which lies on the perpendicular bisector of the line segment joining the points A(-2,-5) and B(2,5) is:

(a) (0,0) (b) (0,2) (c) (2,0) (d) (-2 , 0)

(vi) The third proportion to 2 and 7 is:

(a) $\sqrt{14}$ (b) 24.5 (c) 14 (d) 9

(vii) If sum of n terms of an A.P. is $3n^2$, then the eighth term of the A.P. is:

(a) 25 (b) 29 (c) 54 (d) 45

(viii) What is the value of 'p' for which the line $2x + 3y - 7 = 0$ and $4y - px - 12 = 0$ are perpendicular to each other?

(a) 4 (b) 5 (c) 6 (d) 8

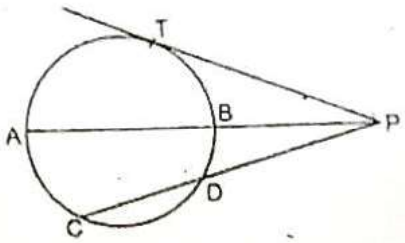
(ix) For a G.P. with 'a' as first term and 'r' as common ratio, its fourth term = m, seventh term = n and tenth term = p.

Assertion (A) : m,n and p are in G.P.

Reasoning (R): $n^2 = (ar^6)^2 = ar^3 \times ar^9 = m \times p$.

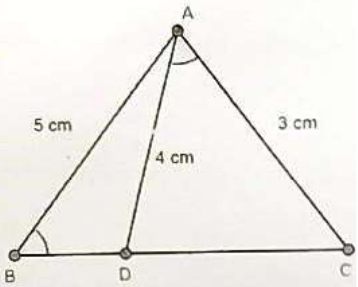
(a) A is true, R is false.
(b) A is false, R is true.
(c) Both A and R are true and R is the correct reason for A.
(d) Both A and R are true and R is the incorrect reason for A.

(x) In the given figure, diameter AB and chord CD of a circle meet at P. PT is a tangent to the circle at T. CD = 7.8 cm, PD = 5cm, PB = 4cm. The length of AB and PT respectively.



(a) 12 cm, 8cm (b) 10 cm, 8 cm (c) 12 cm, 10 cm (d) 8cm, 12 cm

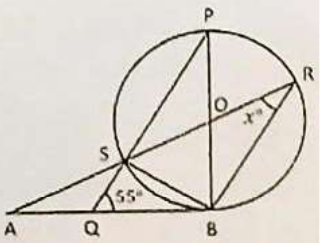
(xi) In the adjoining figure, D is a point on BC such that $\angle ABD = \angle CAD$. If AB = 5 cm, AC = 3cm and AD = 4 cm, then the length of BC is :



(a) 5 cm (b) 3.75cm

(c) 6cm (d) 4.3 cm

(xii) In the given figure, find the measure of x.



(a) 45° (b) 55°

(c) 30° (d) 35°

(xiii) If a sphere and a cube have equal surface areas, then the ratio of the diameter of the sphere to the edge of the cube is:

(a) 1: 2 (b) 2 : 1 (c) $\sqrt{\Pi} : \sqrt{6}$ (d) $\sqrt{6} : \sqrt{\Pi}$

(xiv) In a class test, the marks of scored by 11 students are:

13,17,20,5,3,19,7,6,11,15,17. Find the upper quartile.

9 (b) 17 (c) 11 (d) 20

(xv) If $\sec\theta - \tan\theta = k$, then the value of $\sec\theta + \tan\theta$ is:

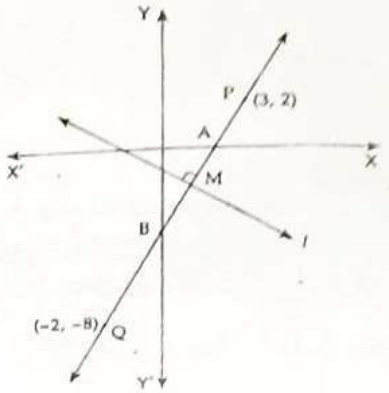
(a) $1 - \frac{1}{k}$ (b) $1 - k$ (c) $1 + k$ (d) $\frac{1}{k}$

Question 2:

(i) In the adjoining figure, line l is perpendicular to PQ. If M is the midpoint of AB, find: [4]

(i) Coordinates pf M.

(ii) Equation of line l.



(ii) Prove that: $\frac{cotA+cosecA-1}{cotA-cosecA+1} = \frac{1+cosA}{sinA}$ [4]

(iii) A man buys ₹50 shares of a company which pays 12% dividend. He buys the shares at such a price that his profit is 15% of his investment. At what price did he buy the shares? [4]

Question 3:

(i) Construct a triangle ABC, having given BC = 7cm, $\angle ABC = 45°$ and median AD = 6cm. [5]

Inscribe a circle in the ΔABC. Measure the radius of the circle.

(ii) Use a graph paper to solve this question: [4]

(a) Plot the points P(0,3), Q(3,-2) and O(0,0).

(b) Plot R, the image of Q, when reflected in the y – axis and write its coordinates.

(c) What is the geometrical name of the figure PQOR.

(d) Also write the equation of the line of symmetry of PQOR.

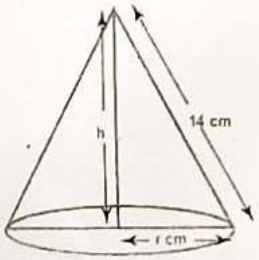
(iii) A semi-circular sheet of metal of diameter 28cm is bent to form an open cone. Find: [4]

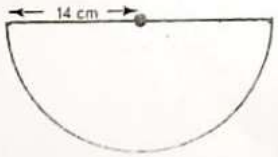
(a) the radius of the cone.

(b) the height of the cone.

(c) the curved surface area of the cone.

(d) the volume of the cone.





SECTION II (20 Marks)
Attempt any 4 questions from this Section

Question 4.

(i) From a cuboidal solid of metallic block of dimensions 15cm x 10cm x 5cm, a cylindrical hole of diameter 7 cm is drilled out. Find the surface area of the remaining block. [3]

(ii) Solve the following equations by using quadratic formula: [3]

$$2\left(\frac{2x-1}{x+3}\right) - 3\left(\frac{x+3}{2x-1}\right) = 5,\ x \neq -3, \frac{1}{2}$$

(iii) The horizontal distance between two towers is 120m. The angles of elevation of the top and angle of depression of the bottom of the first tower as observed from the second tower are 30° and 24° respectively. [4]

Find the height of the two towers.

Question 5:

(i) Find the value of 'k' for which the equation, $2x^2 + kx + 3 = 0$ has equal roots. Also find the roots of the resulting equation. [3]

(ii) The line segment joining the points A(3,2) and B(5,1) is divided at the point P in the ratio 1:2 and it lies on the line $3x - 18y + k = 0$. Find the value of 'k'. [3]

(iii) The marks obtained by 100 students in a Maths test is given below: [4]

Marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90	90-100
No of students.	3	7	12	17	23	14	9	6	5	4

Draw an ogive on a graph sheet and from it determine the:

(i) Median (ii) lower quartile

(iii) number of student who obtained more tham 85% marks in the test.

(iv) Number of students who did not pass in the test if the pass percentage was 35.

Question 6:

(i) Find the coordinates of the point which is three – fourth of the way from A(3,1) to B(-2,5). [3]

(ii) AB and CD are two lines intersecting at 45°. Find the position of a point which is at a distance of 2cm from AB and 1.6cm from CD. [3]

(iii) In the given figure, ΔABC is right angled at B. BDFE is a rectangle. [4]

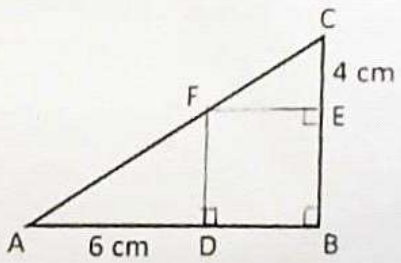
AD= 6cm, CE = 4cm and BC = 12cm.

(a) Prove that ΔADF ~ ΔFEC.

(b) Prove that ΔADF ~ ΔABC.

(c) Find the length of FE.

(d) Find area ΔADF : area ΔABC.



Question 7:

(i) The radii of the internal and external surfaces of a metallic spherical shell are 3cm and 5cm, respectively. It is melted and recast into a solid right circular cylinder of height $10\frac{2}{3}$ cm. Find the diameter of the base of the cylinder. [3]

(ii) The angle of elevation of a stationary cloud from a point 25m above a lake is 30° and the angle of depression of its reflection in the lake is 60°. What is the height of the cloud above that lake-level? [3]

(iii) The weight of 50 apples were recorded as given below. Calculate the mean weight, to the [4]

nearest gram, by the Step Deviation Method.

Weight (in grams)	80 - 85	85 - 90	90 - 95	95 - 100	100 - 105	105 - 110	110 - 115
No of apples	5	8	10	12	8	4	3

Question 8.

(i) Given, $x = \frac{\sqrt{a^2 + b^2} + \sqrt{a^2 - b^2}}{\sqrt{a^2 + b^2} - \sqrt{a^2 - b^2}}$, use Componendo and Dividendo to prove that $b^2 = \frac{2a^2x}{x^2+1}$ [3]

(ii) Find the sum of the first n terms of: [3]

$4 + 44 + 444 + \ldots.$

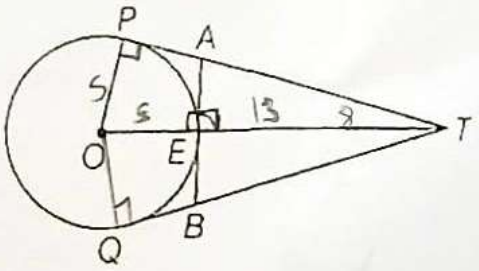
(iii) Graphically, find the mode of the following frequency distribution: [4]

Class	20 - 30	30 - 40	40 - 50	50 - 60	60 - 70	70 - 80
Frequency	4	7	9	11	6	2

Question 9.

(i) [3]

In the given figure, O is the centre of a circle of radius 5 cm, T is a point such that OT = 13cm and OT intersects the circle at E. If AB is the tangent to the circle at E, then find the length of AB.



(ii) By increasing the speed of a car by 10 km/h, the time of journey for a distance of 72 km is reduced by 36 minutes. Find the original speed of the car. [3]

(iii) The sum of first six terms of an A.P. is 42. The ratio of the 10th term to the 30th term is 1 : 3. Calculate the first and the thirteenth term. [4]

****

Question Paper 15

First Preliminary Examination

MATHEMATICS

Grade: X **Max. Marks: 80**

Time Allowed: 3 hours

INSTRUCTIONS:

- **Answers to this paper must be written on the paper provided separately.**
- **You will not be allowed to write during the first 15 minutes.**
- **This time is to be spent in reading the question paper.**
- **The time given at the head of this paper is the time allowed for writing the answers.**
- **The intended marks for the questions or parts of questions are given alongside the questions.**
- **Attempt all questions from Section A and any four question from Section B**
- **All working, including rough work, must be clearly shown and must be done on the same sheet as the rest of the answer. Omission of essential working will result in loss of marks.**
- **Geometrical figures to be constructed wherever applicable.**
- **For geometry, figures are to be copied to the answer script.**
- **This question paper contains 10 printed pages.**

SECTION A

(*Attempt **all** questions from this Section.*)

Question 1 **Choose the correct answers to the questions from the given options. (Do not copy the question, write the correct answers only.)** **[15]**

(i) If the roots of the quadratic equation $x^2 + m(2x + m - 1) + 2 = 0$ are equal, then the value of m is:

(a) 3

(b) 2

(c) 1

(d) -1

(ii) A man deposits ₹x per month for 'y' years in a R.D account. If at the time of maturity, he gets ₹z as interest, then the total maturity amount is:

(a) ₹$(12xy + z)$.

(b) ₹ $12xyz$

(c) ₹ $xy + 12z$

(d) ₹ $\frac{xyz}{12}$

(iii) If a dice is rolled, the ratio of the probability of getting a prime number to the probability of getting a multiple of 3 is:

(a) 1:2

(b) 1:3

(c) 2:3

(d) 3:2

(iv) Which of the following equations represents a line passing through the origin and parallel to the line 3y – 2x + 5 = 0 ?

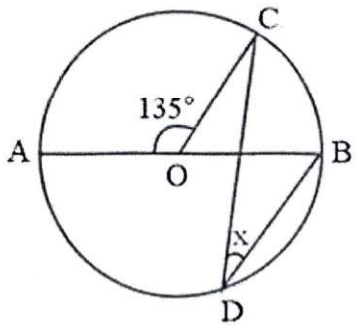
(a) 3y – 2x + 5 = 0

(b) 2x – 3y = 0

(c) 3x – 2y = 0

(d) y = -5

(v) In the given figure, O is the centre of the circle, the value of x is:



(a) 45^0

(b) 135^0

(c) 22.5^0

(d) 40^0

(vi) If $(x + 2)$ is a factor of $3x^3 - x^2 - px - 4$, then the value of p is

(a) 14

(b) 12

(c) 10

(d) 16

(vii) Company P invested ₹12500 in shares and got an annual dividend of ₹500.

Company Q invested ₹13000 in shares and got an annual dividend of ₹650.

Company R invested ₹15000 in shares and got an annual dividend of ₹600.

Which of them has less rate of return on the investment?

(a) Company P

(b) Company Q

(c) Company R

(d) Company P and Company R

(viii) The three vertices of a scalene triangle are always equidistant from a fixed point. The point is:

(a) orthocentre of the triangle

(b) incentre of the triangle

(c) circumcentre of the triangle

(d) centroid of the triangle

(ix) The approximate volume of a human eye is 6.5 cm^3. The volume of a laboratory model of a human eye is 1404 cm^3. The value of k is:

(a) 6

(b) $\frac{1}{6}$

(c) 216

(d) $\frac{1}{216}$

(x) A solid sphere is cut into two identical hemispheres.

Statement 1: The total volume of the two hemispheres is equal to the volume of the original sphere.

Statement 2: The total surface area of the two hemispheres together is equal to the surface area of the original sphere.

(a) Both the statements are true.

(b) Both the statements are false.

(c) Statement 1 is true, and statement 2 is false.

(d) Statement 2 is true, and statement 1 is false.

(xi) If the replacement set is the set of real numbers, the solution of the inequation $\frac{x+3}{8} < \frac{x-3}{5}$ is:

(a) $\{x: x \in R; x > 13\}$.

(b) $\{x: x \in R; x < 13\}$.

(c) $\{x: x \in R; x \geq 13\}$.

(d) $\{x: x \in R; x \leq 13\}$.

(xii) Consider the following frequency table:

Class	0 - 5	6 - 11	12 - 17	18 – 23	24 - 29
Frequency	13	10	15	8	11

The upper limit of the modal class is:

(a) 17

(b) 17.5

(c) 18

(d) 18.5

(xiii) Assertion (A): For a trade X, GST paid is ₹600 and GST collected is ₹720. Therefore, the GST paid by the trader to the State Government is ₹120.

Reason (R): GST deposited with the government = Output Tax – Input Tax = GST collected – GST paid.

(a) A is true, R is false

(b) A is false, R is true

(c) Both A and R are true

(d) Both A and R are false

(xiv) The ratio of the length of a pole and its shadow is, $1:\sqrt{3}$, then the angle of elevation is:

(a) 90^0

(b) 45^0

(c) 30^0

(d) 60^0

(xv) If M = $\begin{bmatrix} x \\ y \end{bmatrix}$ and N = $[p \quad q]$

Assertion (A): Product MN and NM are both possible.

Reason (R) : Matrix multiplication is commutative.

(a) A is true, R is false

(b) A is false, R is true

(c) Both A and R are true

(d) Both A and R are false

Question 2

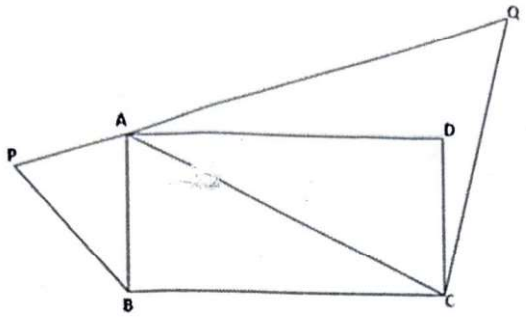
(i) By selling an article for ₹24, a trader loses as much percent as the cost price of the article. Calculate the cost price. **[4]**

(ii) A man opened a recurring deposit account in a branch of Citizen Credit Cooperative Bank. The man deposits certain amount of money per month such that after two years, the interest accumulated is equal to his monthly deposits. **[4]**

(a) Find the rate of interest per annum that the bank was paying for the recurring deposit amount.

(b) If the man deposited ₹500 per month, find the total amount he received after 2 years.

(iii) ABCD is a rectangle where side BC is twice side AB. If $\Delta BAP \sim \Delta ACQ$, find area of ΔBAP: area of ΔACQ. **[4]**



Question 3 (i) Construct an isosceles triangle ABC such that AB = 6cm, BC = AC = 4cm. Bisect angle C internally and mark a point P on this bisector such that CP = 5cm. Find the points Q and R, which are 5 cm from P and also from line AB. **[4]**

(ii) Given $\begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$ M =6 I,where M is a matrix and I is unit matrix of order 2 x 2. **[4]**

(a) State the order of the matrix M.

(b) Find the matrix M.

(iii) A boat is sailing in a straight line from point A(1, 2) to point B(7, 8). A rescue team is stationed at point C(4, 2), which is not on the path of the boat.(A,B an C are non-collinear points) **[5]**

(a) Find the coordinates of the point P on the path AB such that the distance between the rescue team at C and the boat is minimum.

(b) Find the centroid of the triangle formed by the points A, B, and C.

SECTION B

*(Attempt any **four** questions from this Section.)*

Question 4 Solve the following inequation, write down the solution set and represent it on a number line: **[3]**

$-2 + 10x \leq 13x + 10 < 24 + 10x,\ x \in Z$

(ii) The sequence 2, 9, 16..... is given. **[3]**

a) Identify if the given sequence is in AP or GP.

b) Find the difference between the sum of the first 22 and the first 25 terms.

(iii) Prove that: $(1 + tan^2 A)\left(1 + \frac{1}{tan^2 A}\right) = \frac{1}{sin^2 A - sin^4 A}$ **[4]**

Question 5 (i) Using properties of proportion, solve for x. **[3]**

$$\frac{x^4+1}{2x^2} = \frac{17}{8}$$

(ii) If $x^2 + 3x + 2$ is a factor of $3x^3 + x^2 - ax - b$, find the values of a and b. **[3]**

(iii) A restaurant stores ice-cream in a box with a dispenser attached for filling ice-cream cones. The dimensions of the box and the ice-cream cones used by the restaurant are shown in Figure 1 below. To make each serving of dessert, the cone is first filled with ice-cream and then topped with a hemispherical scoop of ice-cream taken from the same box, as shown in Figure 2. Approximately how many desserts can be served out of a completely filled box of ice-cream? **[4]**

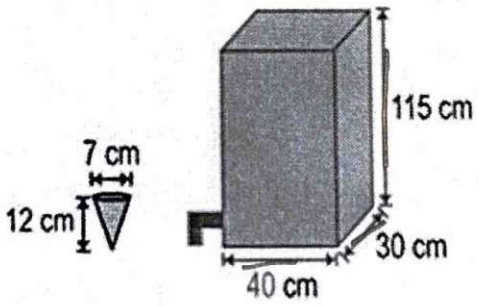


Figure 1



Figure 2

Question 6 (i) Find the mean of the following data using the direct method. Give your answer to the nearest gram. **[3]**

Weight (in gram)	Frequency
80 – 85	5
85 – 90	10
90 – 95	12
95 – 100	12
100 – 105	8
105 – 110	2
110 - 115	1

(ii) If the straight lines $3x - 5y = 7$ and $4x + ay + 9 = 0$ are perpendicular to one another, find the value of a. **[3]**

(iii) The following bill shows the GST rate and the marked price of a few articles: [4]

Sr. No	Item	Quantity	Marked Price (per kg)	GST rate
1	Wheat	5 kg	₹ 35.00	x%
2	Rice	5 kg	₹ 180.00	5%
3	Detergent	y kg	₹ 220.00	18%

Find:

(a) the value of 'x' if the total GST on wheat and rice is ₹ 45,

(b) the value of 'y' if CGST paid for detergent powder is ₹ 39.60,

(c) the total amount to be paid including GST for the above bill.

Question 7 (i) Draw a histogram and estimate the mode for the following frequency distribution: [3]

Class	0 - 10	10 - 20	20 - 30	30- 40	40 - 50
Frequency	5	12	20	9	4

(ii) A man invests ₹10560 in a company, paying 9% dividend, at the time when its ₹100 shares can be bought at a premium of ₹32. Find: [3]

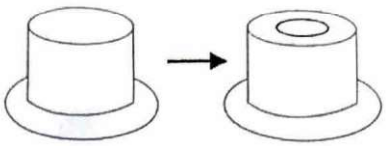
(a) the number of shares bought by him,

(b) his annual income from these shares,

(c) the rate of return on his investment,

(iii) A rectangular piece of cardboard measuring 40 cm by 30 cm is to be made into an open box with a base of 936 cm^2 by cutting equal squares from the four corners and then binding up the sides. Find the length of the side of the square that must be cut from each corner. [4]

Question 8 (i) Shown below is a cake that Subodh is baking for his brother's birthday. The cake is 21 cm tall and has a radius of 15 cm. He wants to surprise his brother by filling gems inside the cake. In order to do that, he removes a cylindrical portion of cake out of the centre as shown. The piece that is removed is 21 cm tall. If the cake weighs 0.5g per cubic cm and the weight of the cake that is left after removing the central portion is 6600 g, find the radius of the central portion that is cut. (Take $\pi=\frac{22}{7}$). [3]



(ii) A game has 8 triangles of which 3 are blue and rest are red and 10 squares of which 6 are blue and rest are red. One piece is lost at random. Find the probability that it is a: **[3]**

(a) triangle

(b) square

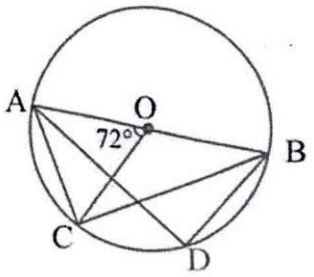
(c) triangle of red colour

(iii) In the figure given below, O is the centre of the circle and AB is the diameter. If AC = BD and $\angle AOC = 72^0$, find: **[4]**

(a) $\angle ABC$

(b) $\angle BAD$

(c) $\angle ABD$



Question 9 (i) Use a graph paper for this question. Take scale: 1cm = 1 unit along both X and Y axis. Plot points A(2, 2) and B(6, -2) in the graph paper and answer the following questions: **[5]**

(a) Reflect point A in the origin to get D and write the co-ordinates of point D.

(b) Reflect point A in the line y= -2 to point C and write the co-ordinates of point C.

(c) Find a point P on CD which is invariant under reflection in x = 0 and write its co-ordinates.

(d) Write the special geometric name of the figure ABCD.

(e) Write down the co-ordinates of point of intersection of diagonals of ABCD.

(ii) From a point P on the ground, the angle of elevation of the top of a 10m tall building is 30^0. A flag is hoisted at the top of the building and the angle of elevation of the top of the flagstaff from P is 45^0. Find the length of the flag and the distance of the building from point P. **[5]**

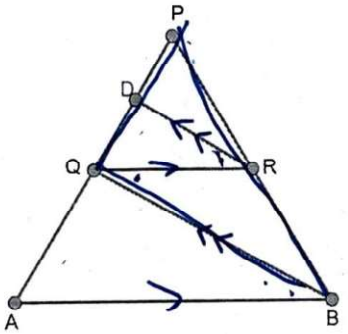
Question 10 (i) The 4th, 6th and the last terms of a GP are 10, 40 and 640 respectively. If the common ratio is positive, find: **[3]**

(a) the first term

(b) common ratio

(c) number of terms

(ii) In the given figure, QR is parallel to AB and DR is parallel to QB. Prove that: $PQ^2 = PD \times PA$ **[3]**



(iii) The table below shows the number of production units (in batches) and the corresponding cost per unit incurred by a company during a particular month: **[4]**

Cost per unit (in ₹)	Number of batches
0 – 50	4
50-100	6
100-150	10
150-200	12
200-250	18
250-300	20
300-350	15
350-400	5

Draw a less than ogive using a suitable scale on graph paper and estimate the median cost per unit.

* * * * *

Question Paper 16

First Preliminary Examination
Subject: Mathematics

Std: 10 **Marks: 80**
Time: 3hrs

INSTRUCTIONS:

- *Answers to this Paper must be written on the paper provided separately.*
- *You will **not** be allowed to write during the first **15** minutes.*
- *This time is to be spent in reading the question paper.*
- *The time given at the head of this Paper is the time allowed for writing the answers.*

- Attempt all questions from Section A and any four questions from Section B.
- All working, including rough work, must be clearly shown, and must be done on the same sheet as the rest of the answer.
- Omission of essential working will result in loss of marks.
- Mathematical tables are provided.

SECTION A

(Attempt all questions from this section)

Question 1 **[15]**

Choose the correct answers to the questionsfrom the given options.
(Do not copy the questions , write the correct answers only.

i)A dealer in a city buys some goods worth Rs 8000 from the same city. If the rate of G.S.T is 15% ,then the dealer pay for the goods bought

(a) Rs 9000 (b) Rs 9200
(b) Rs 9400 (d) Rs 9600

ii) which term of the A.P. 21,42,63,84..... is 210?

(a) 9th *(b) 10th*
(c) 11th *(d) 12th*

iii) The equation of a straight line is 3x – 3y -7 = 0 , the gradient of the line is

(a) $-\frac{7}{3}$ ***(b)*** $\frac{7}{3}$
(c) 1 ***(d)*** 3

iv) If tan (3x + 30) =1, then the value of x is

(a)5^0 (b) 10^0

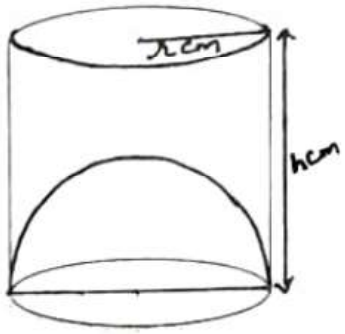
© 15^0 (d) 20^0

v) The value of x, if on dividing $7y^3 + 2y^2 - xy + 14$ by $y + 3$, leaves a remainder -49 is

(a) 18 (b) 36

(c) 24 (d) 48

vi) The capacity of a cylindrical vessel with a hemispherical portion raised upward at the bottom as shown in the figure is



(a) $\frac{1}{3}\pi r^2[2h - 3]$ (b) $\frac{2}{3}\pi r^2[3h - 2r]$

(c) $\frac{1}{3}\pi r^2[3h - 2r]$ (d) *None of these*

vii)Devesh has 450 shares of Rs.200 of a company paying a dividend of 16%, his income after paying an income tax of 20% is

(a) 13550 (b) 12000

(c) 14566 (d) 11520

viii) If (3a + 2b): (5a +3b) = 18: 29 , ~~then 18:29~~ , then a:b is

(a) 7: 3 (b) 4: 3

(c) 8: 5 (d) 7:12

ix)Given that $\begin{bmatrix} x \\ y \end{bmatrix}$ **x A** $= \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.

Statement I : The order of matrix A is 1 x 2.

Statement II : If A of order mxn and B is of order nxp , then AB is of order mxp

(a) Both Statements are true.

(b) Both Statements are false.

(c) Statement I is true and Statement II is false

(d) Statement I is false and Statement II is true.

x) The value of x , which satisfy the inequation $-2\frac{5}{6} < \frac{1}{2} - \frac{2x}{3} \leq 2$,

$\forall$ **x** $\in$ **W is**

(a) {0,1,2,3} (b) { 1,2,5,7}

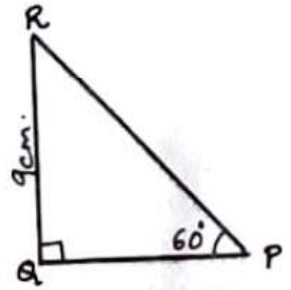
© {0,1,2,3,4} (d) {0}

xi) The equation of a straight line with slope -3 and which intersects X-axis at a distance of 5units to the left of origin is

(a) y-3x+15=0
(b) y-3x-15=0
(c) y+3x+15=0
(d) y+3x-15=0

xii) Statement I : $(\sin^2\theta - 1)(\tan^2\theta + 1) + 1 = 0$
Statement II : $1 + \tan^2\theta = \sec^2\theta$
Which of the following is valid?
(a) Only I
(b) Only II
(c) Both I and II
(d) Neither I nor II

xiii) In the given diagram, the length of PR is



(a) $3\sqrt{3}$ cm
(b) $6\sqrt{3}$ cm
(c) $9\sqrt{3}$ cm
(d) 18cm

xiv) At what price should a 10% of Rs 25 share be quoted when the money is worth 5% ?
(a) 40
(b) 70
(c) 60
(d) 50

xv) If the points A(-5,1) ,B(0,a) and C(8,2) are collinear, then the value of a is
(a) $\frac{18}{13}$
(b) $\frac{13}{18}$
© $\frac{31}{81}$
(d) $\frac{18}{11}$

Question 2

i) If (x+2) and (x+4) are the factors of $3x^3 + ax^2 - 6x - b$, then determine the values of a and b. **[4]**

ii) Prove that: $\frac{\sqrt{SecA-1}}{\sqrt{Sec\,A+1}} + \frac{\sqrt{SecA+1}}{\sqrt{SecA-1}} = 2\text{cosec A}$ **[4]**

iii) Construct a triangle ABC with AB =5.5cm, AC =6cm and angle BAC=105°. Hence: (a) construct the locus of points equidistant from BA and BC. (b) construct the locus of points equidistant from B and C © Mark the point which satisfies the above two loci as P. Measure and write the length of PC. **[4]**

Question 3

i)A spherical glass vessel has a cylindrical neck 8cm long, 2cm in diameter and the diameter of the spherical part is 8.5cm. By measuring the amount of water it holds, a child find its volume to be $345cm^3$. Check whether the child is correct, taking the above as the inside measurements. (Take $\pi = 3.14$) **[4]**

ii) The price of Honda scooty is Rs 62000 including tax (under GST) at the rate of 18% on its marked price. A buyer asks for a discount on the marked price so that after charging GST, the selling price of scooty becomes equal to the marked price. Find the discount amount in which the seller has to allow the deal. **[4]**

iii) Use graph paper for this question,

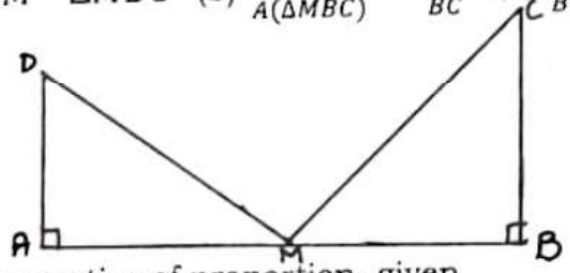
A(1,1) ,B(5,1), C(4,2), D(2,2)are the vertices of the quadrilateral. Name the quadrilateral ABCD . Points A,B,C,D are reflected in the origin onto A', B', C' ,D' , respectively . Locate A', B', C', D', on the graph sheet and write their coordinates. Are D,A, A', and D' collinear points? **[5]**

SECTION B

(Attempt **any four** questions from this section)

Question 4

i) In the given figure ,M is the midpoint of AB, $\angle A = \angle B = 90^{\circ} = \angle CMD$, prove that $(a)\ \Delta DAM \sim \Delta MBC$ (b) $\frac{A(\Delta DAM)}{A(\Delta MBC)} = \frac{AD}{BC}$ (c) $\frac{AD}{BC} = \frac{MD^2}{MC^2}$. **[3]**



ii) Solve for x,using properties of proportion, given

$\frac{x^4+1}{2x^2} = \frac{17}{8}$. **[3]**

iii) Mahesh bought 600 shares of Rs 50 each of 'Excel company' . He sold $1/3^{rd}$ of them when they were at premium of Rs 20 and the remaining when they were at a discount of Rs 5. Find his gain or loss in the whole transaction. **[4]**

Question 5

i) The product of first three terms of a G.P. is 1000.If 6 is added to its second term and 7 is added to its third term , then the term become in AP. Find the (a) first term (b) G.P. **[3]**

ii) Mr.Mukesh opened a recurring account in a bank. He deposited Rs 2500 per month for 2 years. At the time of maturity ,he got Rs 67500. Find the rate of interest per annum. **[3]**

iii) Construct a tangents to a circle of radius 1.8cm from a point on the concentric circle of radius 2.8cm and measure its length. **[4]**

Question 6

i) Rs.480 is divided equally among x children. If the number of children was 20 more than each would have got Rs. 12 less, find the value of x. **[3]**

ii) The equation of a line is $2x - 2\sqrt{3}y - \sqrt{3} = 0$. **[3]**

Find (a) the gradient of the line.

(b) the inclination of the line

(c)the y-intercept of the line.

iii) A right angled triangle with sides 3cm and 4cm is made to revolve about its hypotenuse. Find the volume of the double cone so formed. (Take $\pi = 3.14$). **[4]**

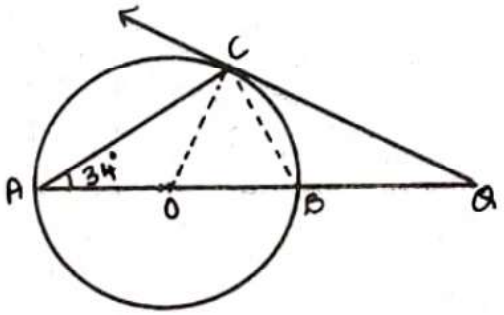
Question 7

i) If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, then find the value of $A^2 + 3A + I$. **[3]**

ii) The line segment joining A(2,3) and B(6,-5) is intercepted by the X-axis at a point K.(a) find the ratio in which K divides AB.

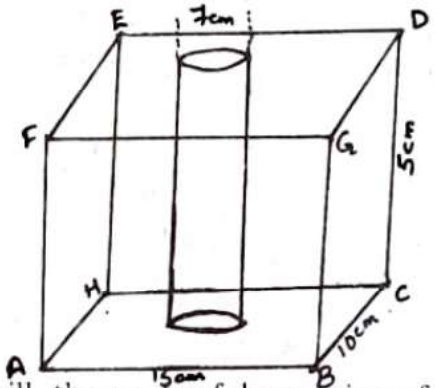
(b) Write the coordinates of the point K **[3]**

iii) In the given figure,AB is a diameter.the tangent at C meet AB produced at Q.if $\angle CAB = 34^\circ$,find $\angle CBA$ and $\angle CQA$. **[4]**



Question 8

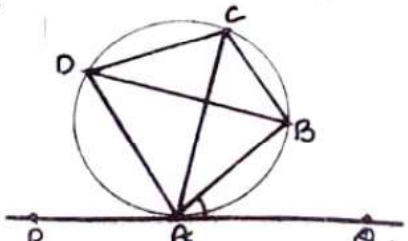
i) In the given figure ,of a cuboidal solid metallic block of dimensions 15cm x10cm x 5cm, a cylindrical hole of diameter 7cm is drilled out. Find the surface area of the remaining block. **[3]**



ii) From the top of a hill, the angle of depression of two consecutive kilometre stone due west are 30° and 45°, respectively. Find the height of the hill. **[3]**

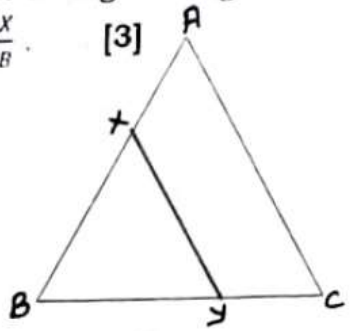
iii) In the given figure ,PQ is a tangent to the circle at A. AB and AD are bisectors of $\angle CAQ$ and $\angle PAC$, respectively.If $\angle BAQ = 30^\circ$, prove that(a) BD is a diameter of the circle. **[4]**

(b) ΔABC is an isosceles triangle.



Question 9

i) In the adjoining figure ,XY//AC and XY divides triangular region ABC into two parts equal in area. Determine $\frac{AX}{AB}$. **[3]**



ii)Solve the quadratic equation $3x^2 - x - 7 = 0$ for x and give your answer correct to 2 decimal places. **[3]**

iii)If the coordinates of the midpoints of the sides of a triangle are (1,1), (2,-3) and (3,4).Find its centroid. **[4]**

Question 10

i) Which is a better investment? 20% at 100 or 25% at 150? **[3]**

ii) Prove $(\sin A + \sec A)^2 + (\cos A + \operatorname{cosec} A)^2 = (1 + \sec A \operatorname{cosec} A)^2$. **[3]**

iii) If $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$, prove that **[4]**

$$\frac{ax-by}{(a+b)(x-y)} + \frac{by-cz}{(b+c)(y-z)} + \frac{cz-ax}{(c+a)(z-x)} = 3.$$

Question Paper 17

FIRST PRELIMINARY EXAMINATION

Class : X | **MATHEMATICS** | **Marks : 80 / Pgs. 8** | **Time : 3 Hrs**

Answers to this paper must be written on the paper provided separately.
You will not be allowed to write in the first 15 minutes.
This time is to be spent in reading the question paper.
The time given at the head of this paper is the time allowed for writing the answers. The intended marks for questions or parts of questions are given in brackets [ ].
Mathematical tables are provided.

SECTION-A
(Attempt all the questions from this section)

Question 1 [15]

i. If $x \geq y$ and $\frac{x}{p} \leq \frac{y}{p}$, then p is
   a) Negative
   b) Positive
   c) Zero
   d) None of these

ii. For what value of m, the lines $5x + 2y = 1$ and $mx + y = 2$ are parallel to each other.
   a) $\frac{1}{5}$
   b) $\frac{2}{5}$
   c) $\frac{3}{5}$
   d) $\frac{5}{2}$

iii. The sum invested to purchase 15 shares of a company of nominal value ₹75 available at a discount of 20% is
   a) ₹60
   b) ₹90
   c) ₹1350
   d) ₹900

iv. Which of the following equation has root -3 ?
   a) $x^2 - x - 12 = 0$
   b) $x^2 + x - 12 = 0$
   c) $x^2 - 7x + 12 = 0$
   d) $x^2 + 2x - 15 = 0$

v. The values of a, b and c from the following transaction of an article within the same city
Listed price = ₹1200
Discount = 20%
Discounted price = a
GST = 18%
CGST = b
SGST = c , are
a) a = ₹960, b = c = ₹86.40
b) a = ₹240, b = c = ₹21.60
c) a = ₹960, b = c = ₹172.80
d) a = ₹240, b = c = ₹43.20

vi. If there are n observations (variates) $x_1, x_2, x_3, \ldots\ldots x_n$ arranged in ascending order then ,
Statement 1: When n is even, median = $\left(\frac{n}{2}\right)^{th}$ observation + $\left(\frac{n+1}{2}\right)^{th}$ observation.
Statement 2: When n is odd , median = $\left(\frac{n+1}{2}\right)^{th}$ observation.
Which of the statement is valid ?
a) Only Statement 1
b) Only Statement 2
c) Both Statement 1 and Statement 2
d) Neither Statement 1 nor Statement 2

vii. If A = $\begin{bmatrix} 2 & 3 \\ 4 & 2 \end{bmatrix}$ and B = $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, then the matrix BA is
a) $\begin{bmatrix} 11 & 16 \\ 10 & 16 \end{bmatrix}$
b) $\begin{bmatrix} 10 & 22 \\ 7 & 17 \end{bmatrix}$
c) $\begin{bmatrix} 11 & 10 \\ 16 & 16 \end{bmatrix}$
d) $\begin{bmatrix} 10 & 7 \\ 22 & 17 \end{bmatrix}$

viii. When a dice is thrown then which of the following probabilities is/are correct ?
Statement 1 :The probability of getting an even number is $\frac{1}{2}$.
Statement 2 :The probability of getting a prime number is $\frac{1}{2}$.
Statement 3 :The probability of getting a number less than 5 is $\frac{1}{2}$.
a) Only 1 and 2 are correct
b) Only 2 and 3 are correct
c) Only 3 and 1 are correct
d) All 1,2 and 3 are correct

ix. The number of the terms in the Arithmetic Progression 18, $15\frac{1}{2}$, 13 .... , −47 is

a) 27
b) 28
c) 26
d) 25

x. **Assertion:** The value of $\sqrt{\frac{1+cosA}{1-cosA}} = cosecA + cotA$

Reason : For any angle A, where $0° \leq A \leq 90°$, the equation $sin^2A + cos^2A = 1$, is always true.

Select the correct answer from the following

a) Both assertion and reason are correct and reason is the correct explanation of assertion.
b) Both assertion and reason are correct but reason is not the correct explanation of assertion.
c) Assertion is correct but reason is incorrect.
d) Assertion is incorrect but reason is correct.

xi. The ratio of the heights of two right circular cones is 3:4 and that of their base radii is 4:5. Find the ratio of their volumes.

a) 16:25
b) 12:25
c) 4:5
d) 3:4

xii. The length of a tangent from an external point P on a circle with centre 'O' is

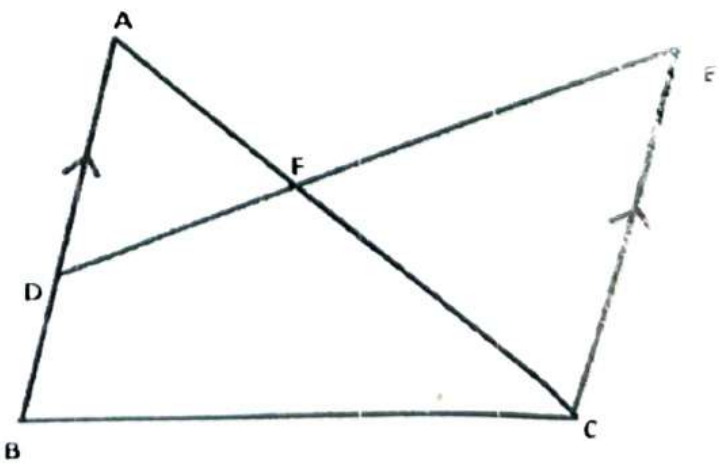
a) Always greater than OP
b) Always equal to OP
c) Always less than OP
d) Can't be estimated

xiii. Naveen Deposits ₹800 every month in a recurring deposit account for 6 months. if he receives ₹4884 at the time of maturity , then the interest he earns

a) ₹84
b) ₹42
c) ₹24
d) ₹284

xiv. In the following figure which of the two triangles are similar?

a) $\Delta ADF \sim \Delta CEF$
b) $\Delta ABC \sim \Delta CEF$
c) $\Delta ADF \sim \Delta ABC$
d) None of these



xv. **Assertion**: $2x^3 + 3x^2 - 4x + 2$ is a polynomial of degree 2.
Reason : The highest power of the variable x in a given polynomial is the degree of the polynomial.
Select the correct answer from the following .

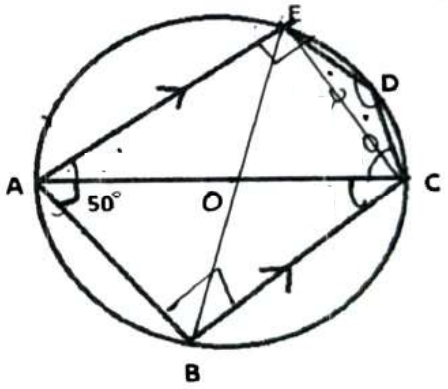
a) Both assertion and reason are correct and reason is the correct explanation of assertion.
b) Both assertion and reason are correct but reason is not the correct explanation of assertion.
c) Assertion is correct but reason is incorrect.
d) Assertion is incorrect but reason is correct.

Question 2

i. In an Arithmetic Progression the 4th and 6th terms are 8 and 14 respectively, find
   a) The first term
   b) The common difference
   c) Sum of the first 20 terms [4]

ii. A tent is in the shape of a cylinder surmounted by a conical top. If the height and radius of the cylindrical part are 7m each and the total height of the tent is 14m. Find: [4]
   a) The quantity of air contained inside the tent .
   b) The radius of a sphere whose volume is equal to the quantity of air inside the tent.

iii. In the given figure ABCDE is a pentagon inscribed in a circle such that AC is the diameter and side BC ∥ AE, if ∠BAC = 50°, find [4]
   a) ∠ACB
   b) ∠EDC
   c) ∠BEC



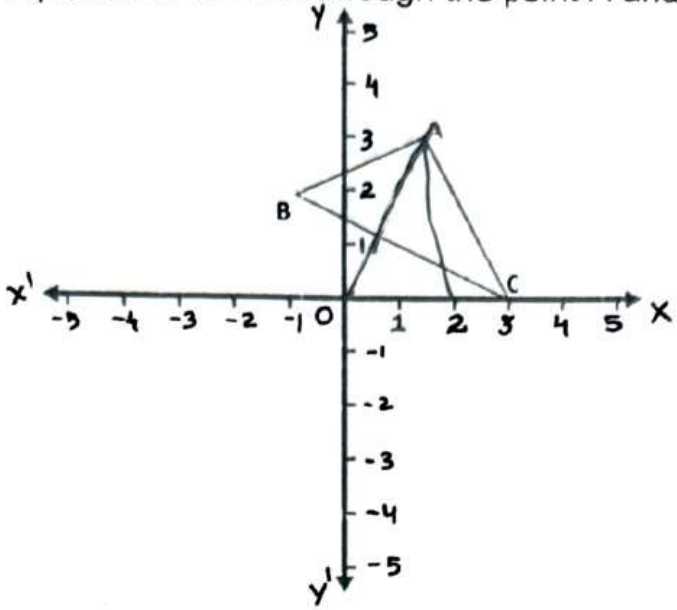
Question 3

i. While factorising a given polynomial using remainder and factor theorem a student finds that (2x+1) is a factor of $2x^3 + 7x^2 + 2x - 3$ [4]
   a) Is the student's solution correct stating that (2x+1) is a factor of the given polynomial .Give a valid reason for your answer.
   b) Also ,factorise the given polynomial completely.

ii. In the given figure write. [4]

a) The co-ordinates of the points A, B and C.

b) The equation of the line through the point A and parallel to BC.



ii. Use graph paper for this question. [5]

(Take 1 cm = 1 unit on both x and y axes)

a) Plot the following points on your graph sheet.
A(-4,0) , B(-3,2) ,C(0,4) ,D(4,1) and E(7,3)

b) Reflect the points B , C ,D and E on the x-axis and name them as B' ,C' ,D' and E' respectively also write their coordinates.

c) Join the points A,B,C,D,E,E',D',C',B' and A in order.

d) Name the closed figure formed.

SECTION-B

(Attempt any 4 questions from the following)

Question 4

i. Draw a Histogram from the following frequency distribution and find the mode from th graph. [3]

Class	0-5	5-10	10-15	15-20	20-25	25-30
Frequency	2	5	18	14	8	5

Solve the following quadratic equation and find the value of x. Give your answer correct to 2 decimal places. $x^2 - 4x - 8 = 0$. [3]

If the pth,q^{th} and r^{th} terms of G.P are a,b and c respectively, prove that $a^{q-r}.b^{r-p}.c^{p-q} = 1$ [4]

Question 5

Each of the letter of the word "HOUSEWARMING" is written on the cards and put in a bag. If a card is drawn at random from the bag after shuffling, what is the probability that the letter on the card is

a) A vowel.
b) One of the letters of the word SEWING.
c) Not a letter from the word WEAR . [3]

ii. Using properties of proportion, solve for x .

$$\frac{\sqrt{3x+4}+\sqrt{3x-5}}{\sqrt{3x+4}-\sqrt{3x-5}} = 9$$ [3]

iii. A buoy is made in the form of a hemisphere surmounted by a right cone whose circular base coincides with the plane surface of hemisphere. The radius of the base of the cone is 3.5meters and its volume is two-third of the hemisphere. Calculate the height of the cone and the surface area of buoy correct to two decimal places. [4]

Question 6

i. A company with 500 shares of nominal value ₹120 declares an annual dividend of 15% calculate: [3]

a) The total amount of dividend paid by the company.
b) Annual income of Mr.Sharma who holds 80 shares of the company.
c) If the return percent of Mr. Sharma from his shares is 10%, find the market value of each share.

ii. Solve the given linear inequation and represent the solution set on the number line.

$-\frac{8}{3} \leq x+\frac{1}{3} < 3+\frac{1}{3}, x \epsilon R$ [3]

iii. Prove the following identity [4]

$$\sqrt{\frac{1+cosA}{1-cosA}} = \frac{tanA+sinA}{tanA.sinA}$$

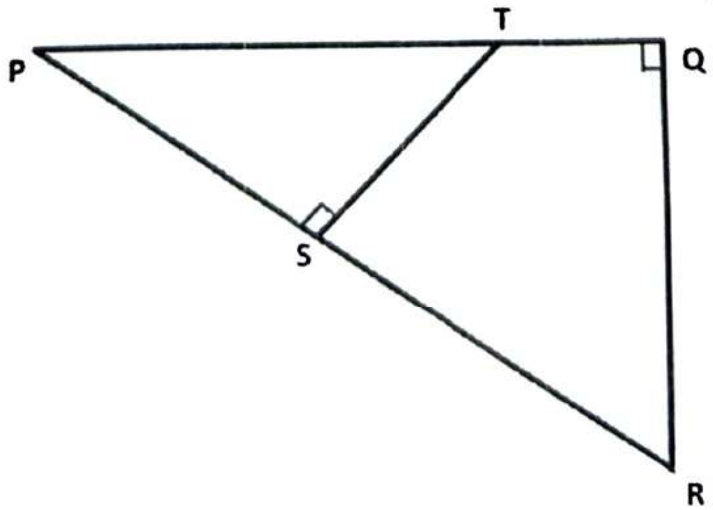
Question 7

i. A man has a recurring deposit account in a bank for $3\frac{1}{2}$ years. If the rate of interest is 12% per annum and the man gets ₹10,206 on maturity, find the value of monthly installment. [3]

200

ii. In the given figure ,∠PQR = ∠PST = 90°, PQ = 5cm and PS = 2 cm. [3]

a) Prove that ΔPQR ~ ΔPST.
b) Find Area of ΔPQR : Area of quadrilateral SRQT.



iii . Find the mean of the following distribution by short-cut method. [4]

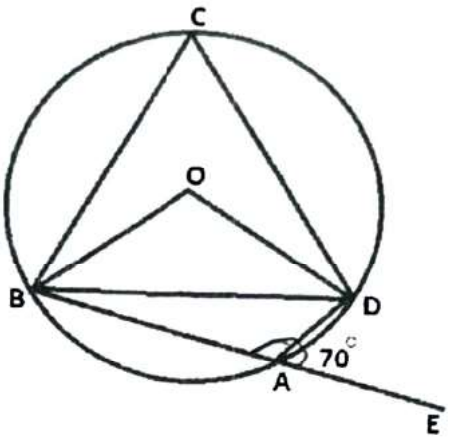
Class Interval	20-30	30-40	40-50	50-60	60-70	70-80
Frequency	10	6	8	12	5	9

Question 8

i. Find the equation of line through the point of intersection of the lines $2x + 5y = 9$ and $5x - 2y = 8$ and perpendicular to the line $4x + 3y = 7$. [3]

ii. In the given figure O is the centre of the circle , $\angle DAE = 70°$. Find [3]

a) $\angle BCD$
b) $\angle BOD$
c) $\angle OBD$



iii. Mrs Arora bought the following articles from a departmental store [4]

S. No	Item	Price	GST	Discount
1.	Hair Oil	₹1200	18%	₹100
2.	Cashew nuts	₹600	12%	-

Find

a) Total GST paid.
b) Total bill amount including GST.

Question 9

i. A man on the deck of a ship is 10 m above water level,He observes that the angle of elevation of the top of a cliff is 42° and the angle of depression of the base is 20°. Calculate the distance of the cliff from the ship and the height of the cliff. [5]

ii. The marks obtained by 100 students in a mathematics test are given below.

Marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90	90-100
No. of students	3	7	12	17	23	14	9	6	5	4

Draw an ogive for the given distribution on a graph sheet.
Use ogive to estimate [5]

a) Median
b) The number of students who obtained more than 85% marks in the test.
c) The number of students who did not pass in the test if the pass % was 35.

Question 10

i. An aeroplane travelled a distance of 400 kilometres at an average speed of x kilometre per hour. On the return journey the speed was increased by 40 kilometre per hour.Write down the expression for the time taken for

a) The onward journey

b) The return journey

if the return journey took 30 minutes less than the onward journey ,write down an equation in x and find its value. [3]

ii. Given $\begin{bmatrix} 8 & -2 \\ 1 & 4 \end{bmatrix} X = \begin{bmatrix} 12 \\ 10 \end{bmatrix}$. Write down [3]

a) The order of matrix X.

b) The matrix X.

iii. By using a ruler and compass ,construct a ΔABC with BC = 6.5 cm , AB = 5.5 cm, AC = 5 cm. Construct the incircle of the triangle. Measure and record the radius of the incircle. [4]

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Question Paper 18

FIRST PRELIMINARY EXAMINATION -

MATHEMATICS

CLASS: X **Marks: 80**

Time: 3 Hrs

Answer to this paper must be written on the paper provided separately.

You will not be allowed to write during first fifteen minutes.

This time is to be spent in reading the question paper.

The time given at the head of this paper is the time allotted for writing the answers.

Attempt **all** questions from **section A** and **any four** from **section B.**

All working, including rough work, must be clearly shown and must be done on the same sheet as the rest of the answer.

Omission of essential working will result in loss of marks.

The intended marks for questions or parts of questions are given in brackets [ ].

Mathematical tables are provided.

SECTION A (40 Marks)

Attempt all questions from this section.

Question 1: Choose the correct answers to the questions from the given options: [15x1=15]

i. If 3 is a root of quadratic equation $x^2 - px + 3 = 0$, then p is equal to:

(a) 4 (b) 3 (c) 5 (d) 2

ii. If $\begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ -8 \end{bmatrix}$, then the values of x and y respectively, are:

(a) 1,-2 (b) -2,1 (c) 1,2 (d) –2,-1

iii. A letter is chosen at random from all the letters of English alphabet. The probability that the letter chosen is a vowel, is:

(a) $\frac{4}{26}$ (b) $\frac{5}{26}$ (c) $\frac{21}{26}$ (d) $\frac{5}{24}$

iv. Mohit opened a recurring deposit account in a bank for 2 years. He deposits ₹1000 every month and receives ₹25,500 of maturity. The interest he earned in 2 years is :

(a) ₹13500 (b) ₹3000 (c) ₹24000 (d) ₹1500

v. What number must be subtracted from the polynomial, if (2x-1) is a factor of $2x^3 - x^2 + 3x$?

(a) $-\frac{3}{2}$ (b) $\frac{5}{2}$ (c) $\frac{3}{2}$ (d)None of these.

vi. If two lines $kx - 2y - 7 = 0$ and $8x - ky + 6 = 0$ are parallel to each other, then the value of k is:

(a) ±2 (b) ±4 (c) ±6 (d) ±8

vii. If $2 \leq 2x - 3 < 5, x \epsilon R$ then the solution set is

(a) $\{x: x \epsilon R, \frac{5}{2} \leq x < 4\}$ (b) $\{x: x \epsilon R, \frac{5}{2} \leq x \leq 4\}$ (c) {3,4} (d) None of these

viii. The mean of ten numbers is 58. If one of the numbers is 40, then the mean of other nine numbers is:

(a) 18 (b) 60 (c)162 (d) 540

ix. In the adjoining figure, if O is the centre of the circle then the value of x is



(a) 18 (b) 20 (c) 24 (d) 30

x. **Assertion (A)**: The value of $(\cos^2 A - 1)\operatorname{cosec}^2 A = 1$

Reason(R): $1 - \sin^2 A = \cos^2 A$

(a) A is true, R is false. (b) A is false, R is true.

(c) Both A and R are true. (d) Both A and R are false

xi. A right angled triangle has sides 5cm, 12cmand 13cm.

Statement 1: If the triangle is resolved about the side 12cm a cylinder is formed.

Statement 2: Id the triangle is resolved about the side 13cm a double cone is formed.

(a) Both the statements are true. (b Both the statements are false.)

(c) Statement 1 is true, statement 2 is false. (d)Statement 1 is false, statement 2 is true.

xii. If $\frac{a^2+b^2}{a^2-b^2} = \frac{37}{12}$ *then* $a : b$ *is*

(a) 7:5 (b)5:7 (c) 5:8 (d) 8:3

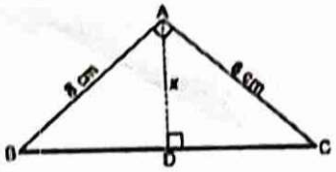
xiii. ₹ 25 shares of company are selling at ₹ 20. If the company is paying a dividend of 12%, then the rate of return is:

(a) 10% (b) 18 % (c) 15% (d) 12%

If (5 m + 2), (4 m – 1) and (m + 2) are in AP, then the value of m is:

(a) 1 (b) 2 (c) –1 (d) 3

In the given figure, if $\angle BAC = \angle ADB = 90°$, then the value of x is :



(a) 6 cm (b) 7 cm (c) 4.8 cm (d)5.2 cm

Question 2

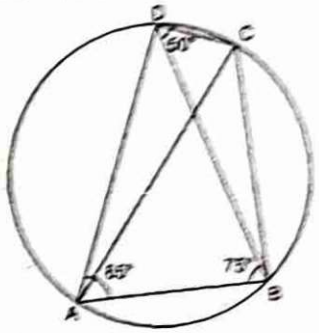
(i) Mrs. Roy went to departmental store and bought the following items:

Sr. No	Item	Price (₹)	Quantity	Rate of G.S.T
1	Walnut	2700 per packet	2 packets	5%
2	Atta	240 per kg	20 kg	0%
3	Butter	200 per packet	5 packets	12%

Find: a: Total GST paid by Mrs. Roy b: Total amount paid by Mrs. Roy [4]

(ii) Show that $2x+7$ is a factor of $2x^3 + 5x^2 - 11x - 14$. Hence, factorise the given expression completely, using the factor theorem. [4]

(iii) In the given figure, $\angle BAD=65°$, $\angle ABD = 75°$, $\angle BDC=50°$. Find (i)$\angle BCD$, (ii)$\angle ADB$. Hence, show that AC is a diameter. [4]



Question 3

(i) A gulab jamun, contains sugar syrup up to about 30% of its volume. Find approximately how much syrup would be found in 45 gulab jamuns, each shaped like a cylinder with two hemispherical ends with length 5 cm and diameter 2.8cm [4]



(ii) Prove that $\frac{sinA}{1+cotA} - \frac{cosA}{1+tanA} = sinA - cosA$ [4]

(iii) A life insurance agent found the following data for distribution of ages of 100 policy holders:

Ages in years	20-25	25-30	30-35	35-40	40-45	45-50	50-55	55-60
Policy Holders	2	4	12	20	28	22	8	4

On a graph sheet draw an ogive using the given data. Take 2cm = 5years along one axis and 2cm =10 policy holder along the other axis. Use your graph to find:

a. The median age.

b. Number of policy holder whose age is above 52 years. [5]

SECTION B

(Attempt any four questions from this Section.)

Question 4

(i) Solve the equation $x^2 - 5x - 5 = 0$ and give your answer correct to two decimal places. [3]

(ii) Draw a histogram to represent the following data and find mode. [5]

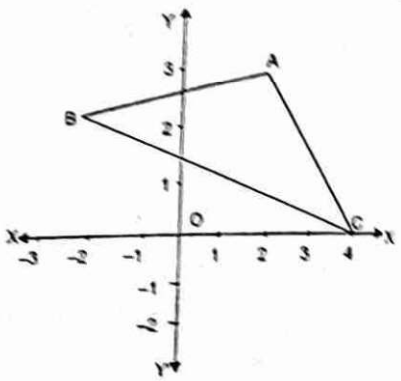
Pocket money in ₹	No. Of students
150-250	5
250-350	10
350-450	7
450-550	4
550-650	6
650-750	2

[4]

(iii). In the given figure, write

a) The co-ordinates of A, B and C

b) the equation of the line through A and parallel to BC



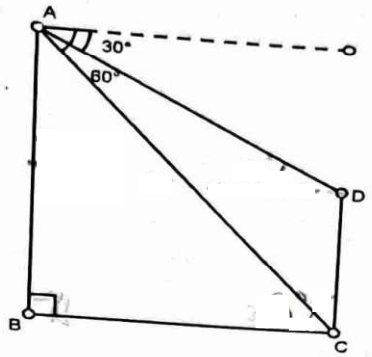
Question 5

(i) The age of a father is equal to the square of the age of his son. The sum of the age of the father and five times the age of the son is 66 years. Find their present ages. [3]

(ii) Rohit invests ₹9000 in a company paying a dividend of 6% per annum. Each share has a face value of ₹100 and is purchased at ₹150. What is his annual income from the dividend? 2. He sells 50% of his shares when the price rises to ₹200. What is his gain on this transaction? [3]

(iii) In the given figure, from the top of a building AB = 60 m high, the angles of depression of the top and bottom of a vertical lamp post CD are observed to be 30° and 60° respectively. Find:

(a) the horizontal distance between AB and CD.
(b) the height of the lamp post. [4]



Question 6

(i) Prove that: $sinA(1+tanA)+cosA(1+cotA)=cosecA+secA$ [3]

(ii) Solve the following Inequation and write the solution set and represent it on number line.

$$-3(x-7)\geq 15-7x>\frac{x+1}{3}\ ,x\in R$$ [3]

(iii) Given that $\frac{a^3+3ab^2}{b^3+3a^2b}=\frac{63}{62}$. using property find $a:b$ [4]

Question 7

(i) A hollow sphere of 6 cm and 8 cm internal and external radii respectively is melted and recast into small cones of 2 cm base radius and 8 cm height. Find the number of cones. [3]

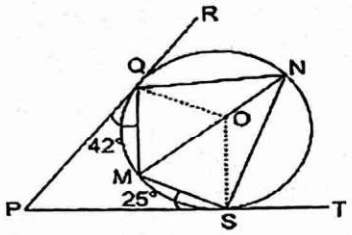
(iii) The 1st term, the last term and the common difference of an arithmetic progression are 98, 1001 and 7. Find the following for the given A P.
(a) number of terms 'n'.
(b) sum of 'n' terms. [3]

(iii) Use ruler and compasses only for the following question. All construction lines and arcs must be clearly shown.
(i) Construct a △ABC in which BC = 6.5 cm, ∠ABC = 60° and AB = 5 cm.
(ii) Construct the locus of points at a distance of 3.5 cm from A.
(iii) Construct the locus of points equidistant from AC and BC.
(iv) Mark 2 points X and Y which are at a distance of 3.5 cm from A and also equidistant from AC and BC. Measure XY. [4]

Question 8

(i) A man on the top of a lighthouse observes the angle of depression of two ships on the opposites sides of the light house as 30° and 50 ° respectively. If the height of the lighthouse is 80 m, find the distance between the two ships. Give your answer to the nearest metre. (Draw the necessary diagram to the question) [5]

(ii) In the given diagram, O is the centre of the circle. PR and PT are two tangents drawn from the external point P and touching the circle at Q and S respectively. MN is a diameter of the circle. Given ∠PQM = 42° and ∠PSM = 25°. Find
(i) ∠OQM (ii) ∠QNS (iii) ∠QOS (iv) ∠QMS



[5]

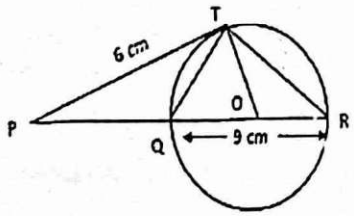
Question 9

(i) $A=\begin{bmatrix}3 & -2\\ -1 & 4\end{bmatrix}, B=\begin{bmatrix}6\\ 1\end{bmatrix}$ and $C=\begin{bmatrix}-4\\ 5\end{bmatrix}$ evaluate $AB-5C$ [3]

(ii) In the given figure, O is the centre of diameter RQ peoduced at P [3]

(a) *Prove* $\Delta PQT \sim \Delta PTR$

(b) *If* $PT = 6\ cm, QR = 9\ cm$. *Find the length of* PQ



(iii) The following table gives the marks scored by students in examination: Calculate mean marks with step deviation method correct to 2 decimal places. [4]

Marks	0-5	5-10	10-15	15-20	20-25	25-30	30-35	35-40
No.of students	3	7	15	24	16	8	5	2

Question 10

(i) A model of a ship is made to a scale of 1:200. [4]
(a) The length of the model is 4 m. Calculate the length of the ship.
(b) The area of deck of the ship is 1,60,000 m^2. Find the area of the deck of the model.
(c) The volume of the model is 200 litres. Calculate the volume of the ship in m^3.

(ii) The 1st and the 8th terms of GP are 4 and 512 respectively. Find [3]
(a) The common ratio
(b) The sum of its first 5 terms

(iii) Sixteen cards are labelled as a, b, c,…, m, n, o, p. They are put in a box and shuffled. A boy is asked to draw a card from the box. What is the probability that the card drawn is:
(i) A vowel
(ii) A consonant
(iii) None of the letter of the word median. [3]

Question Paper 19

BOOSTER EXAMINATION
MATHEMATICS

Time allowed: Three hours

Class: X

Marks: 80

Answers to this paper must be written on the paper provided separately.
*You will **not** be allowed to write during the **first 15 minutes.***
This time is to be spent in reading the question paper.
The time given at the head of this paper is the time allowed for writing the answers.

*Attempt **all** questions from **Section A** and **any four** questions from **Section B**.*
All working, including rough work, must be clearly shown and must be done on the same sheet as the rest of the answer.
Omission of essential working will result in loss of marks.
The intended marks for questions or parts of questions are given in brackets [ ].
Mathematical tables are provided.

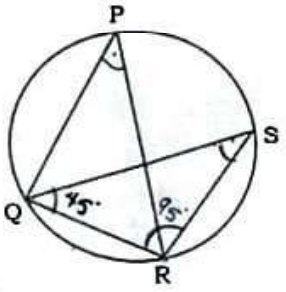
Section A
*(Attempt **all** questions from this **Section**.)*

Question 1

Choose the correct answers to the questions from the given options. **[15]**
(Do not copy the question, write the correct answers only.)

(i) The sum of money required to buy 50, ₹40 shares at ₹38.50 is
(a) ₹1920 (b) ₹1925 (c) ~~₹1925~~ (d) ₹1952

(ii) In the adjoining figure, $\angle SQR = 45°$ and $\angle QRS = 95°$, then $m\angle QPR$ is :
(a) 40° (b) 45°
(c) 30° (d) 50°



(iii) Which of the following equation represents a line equally inclined to the axes?
(a) $2x - 3y + 7 = 0$ (b) $x - y = 7$
(c) $x = 7$ (d) $y = -7$

(iv) If A = [3 −2] and B = $\begin{bmatrix} -1 & 4 \\ 2 & 0 \end{bmatrix}$

Assertion(A): Product of AB of the two matrices A and B is possible
Reason(R): Number of columns of matrix A is equal to number of rows in matrix B
(a) A is true, R is false.
(b) A is false, R is true.
(c) Both A and R are true and R is the correct reason for A.
(d) Both A and R are true and R is the incorrect reason for A.

(v) If the cost of the article is ₹25000and CGST paid by the owner is ₹2250, the GST is

(a) 18% (b) 15% (c) 9% (d) 10%

(vi) A ladder is leaning against a vertical pole. The ladder is 10m long and forms an angle of 45° with the ground. How far up the wall does the ladder reach?

(a) $5\sqrt{3}$ (b) $5\sqrt{2}$ (c) $\sqrt{2}$ (d) $2\sqrt{2}$

(vii) If 3 is the root of quadratic equation $x^2 - px + 3 = 0$ then p is equal to :

(a) 2 (b) 3 (c) 4 (d) 5

(viii) $\tan\theta\sqrt{1-sin^2\theta}$ is equal to

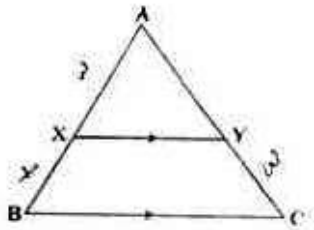
(a) $\cos\theta$ (b) $\sin\theta$ (c) $\text{cosec}\,\theta$ (d) $\sec\theta$

(ix) In a cylinder, if the radius is halved and the height is doubled then the volume will be:

(a) same (b) doubled (c) halved (d) four times

(x) In the adjoining figure $XY \parallel BC$, AX = 3cm, BX = 4cm, CY = 3cm, find the length of AY

(a) 2cm (b) 2.25cm (c) 3.5cm d) 4cm



(xi) Which of the following is not a geometric progression?

(a) $\frac{1}{3}$,1,3,9 (b) $\frac{1}{5},\frac{1}{5},\frac{1}{5},\frac{1}{5}$ (c) -2,4,-8,16 (d) 2,0,4,0,8

(xii) A solid sphere is cut into two identical hemispheres.

Statement 1: The total volume of the two hemispheres is equal to the volume of the original sphere.

Statement 2: The total surface area of the two hemispheres together is equal to the surface area of the original sphere.

Which of the following is valid?

(a) Both the statements are true

(b) Both the statements are false

(c) Statement 1 is true and statement 2 is false

(d) Statement 1 is false and statement 2 is true.

(xiii) Given that sum of the squares of the first seven natural numbers is 140, then their mean is:

(a) 20 (b)70 (c)280 (d) 980

(xiv) Given $x + 2 \leq \frac{x}{3} + 3$ *and x is a* prime number. The solution set for x is

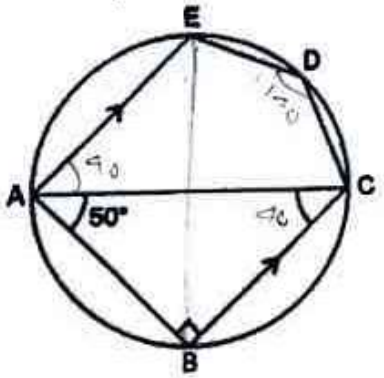
(a) ∅ (b) {0} (c) {1} (d) {0, 1}

Assertion(A): If a chord and tangent intersect externally, then the product of the lengths of the segments of the chord is equal to the square of the length of the tangent from the point of contact to the point of intersection.
Reason (R): A tangent touches the circle at exactly one point.
(a) A is true, R is false.
(b) A is false, R is true.
(c) Both A and R are true and R is the correct reason for A.
(d) Both A and R are true and R is the incorrect reason for A.

Question 2

(i) A tent is in the shape of a cylinder surmounted by a conical top. If the height and the radius of the cylindrical part are 7m each and the total height of the tent is 14m. Find the: **[4]**
(a) quantity of air contained inside the tent.
(b) radius of a sphere whose volume is equal to the quantity of air inside the tent.

(ii) In the given figure, ABCDE is a pentagon inscribed in a circle such that AC is the diameter and side BC ∥ AE. If ∠BAC = 50°, find giving reasons: **[4]**
(a) ∠ACB
(b) ∠EDC
(c) ∠BEC
(d) Hence prove that BE is also a diameter.



(iii) Prove that: $\left(\frac{1-\tan\theta}{1-\cot\theta}\right)^2 = tan^2\theta$ **[4]**

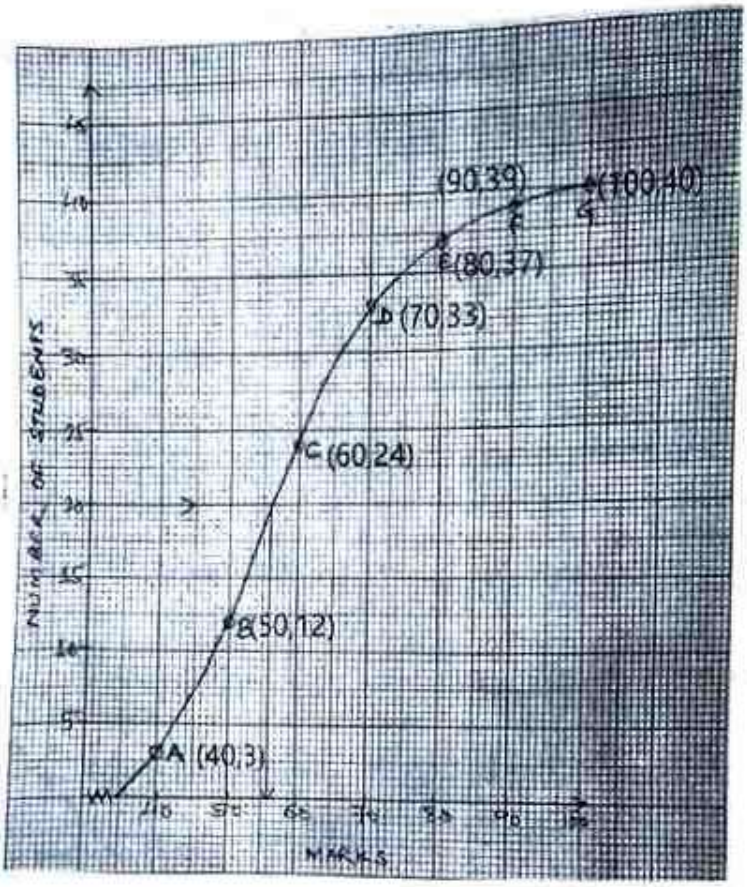
Question 3

(i) Rohit invested Rs. 9600 on ₹100 shares at ₹ 20 premium paying 8% dividend. Rohit sold the shares when the price rose to ₹160. He invested the proceeds in 10% ₹ 50 shares at ₹ 40. Find the **[4]**
(a) original number of shares
(b) sale proceeds
(c) new number of shares
(d) Change in two dividends

(ii) The angle of elevation of a jet plane from a point A on the ground is 60°. After a flight of 15 seconds the angle of elevation changes to 30°. If the jet is flying at a constant height of $1500\sqrt{3}$ m, find the speed of the jet plane in km/hr. **[4]**

(iii) Study the graph and answer each of the following: **[5]**
(a) Name the curve plotted
(b) Total number of students
(c) The median marks

(d) Number of students scoring between 50 and 80 marks



Section B

*(Attempt **any four** questions from this **Section.**)*

Question 4

(i) The first and the last term of a Geometrical Progression are 3 and 96 respectively. If the common ratio is 2, find: **[3]**

(a) n, the number of terms of the G.P.

(b) Sum of the n terms

(ii) Solve the following inequation and represent the solution set on the number line: **[3]**

$$-3 < -\frac{1}{2} - \frac{2x}{3} \leq \frac{5}{6}, x \in R.$$

(iii) A dealer buys a washing machine from a wholesaler for ₹ 30000. He marks the price of the washing machine 10% above the cost price and sells to a customer at a discount of 5% on the marked price if the sale is intra state and the rate of GST is 12%. Find: **[4]**

(a) the marked price of the washing machine.

(b) the amount customer pays including the tax.

(c) the amount of tax paid by dealer to the Central Government.

(d) the amount of tax received by the State Government.

estion 5

Solve the following quadratic equation and give your answer correct to two decimal places: $4x^2 - 5x - 3 = 0$. **[3]**

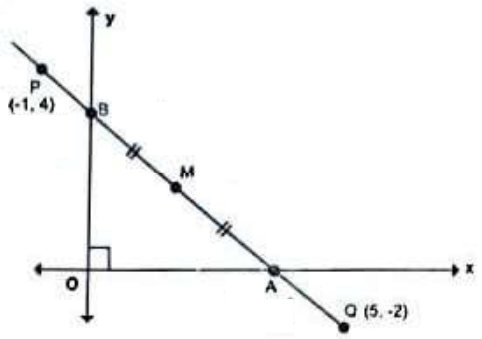
(ii) Using ruler and compass construct a ΔABC where AB = 3cm, BC = 4cm and $\angle ABC = 90°$. Hence construct a circle circumscribing ΔABC. Measure and write down the radius of the circle. **[3]**

(iii) A straight line passes through the points P (-1, 4) and Q (5, -2). It intersects the co-ordinates axes at points A and B. M is the midpoint of the segment AB. Find **[4]**

(a) Equation of the line

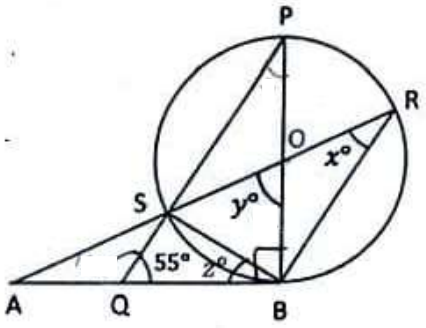
(b) the co-ordinates of points A and B

(c) the co-ordinates of points M



Question 6

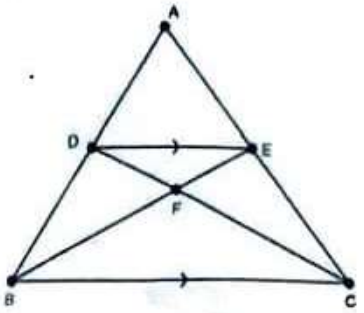
(i) In the given figure, O is the center of the circle and AB is a tangent to the circle at B. If $\angle PQB = 55°$. Find the value of the angles x, y and z. **[3]**



(ii) Calculate the mean of the following data using short cut method: **[3]**

C.I.	0- 10	10 - 20	20 – 30	30 - 40	40 – 50
Frequency	6	8	10	2	4

(iii) In the given figure, DE is parallel to BC and $\frac{AD}{DB} = \frac{3}{2}$,



(a) Determine the ratios of $\frac{AD}{AB}$ and $\frac{DE}{BC}$

(b) What is the ratio of areas of ΔDFE and ΔBFC ?

[4]

Question 7:

(i) Evaluate: $\begin{bmatrix} 4\cos 30° & 2\cos 60° \\ \sin 90° & 2\cos 0° \end{bmatrix} \begin{bmatrix} 4 & 5 \\ 5 & 4 \end{bmatrix}$ **[3]**

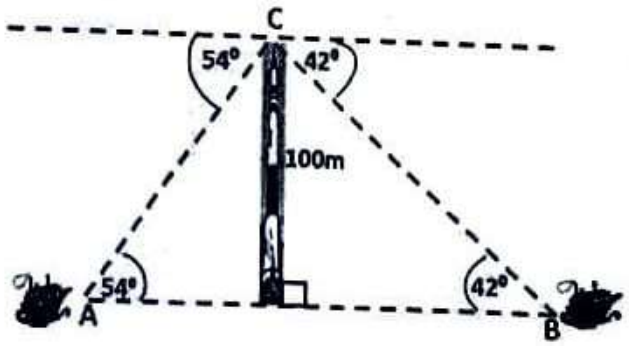
(ii) A man invested ₹ 29040 in 15% ₹100 shares at a premium of 20%. Calculate: **[3]**

(a) the number of shares he buys.

(b) his yearly dividend.

(c) the percentage return on his investment.

(iii) The angles of depression of two ships A and B on the opposite sides of a lighthouse of height 100m are respectively 54° and 42°. The line joining the two ships passes through the foot of the lighthouse. Find the distance between the two ships A and B. Give your answer correct to nearest whole number. **[4]**



Question 8:

(i) Find the equation of a line that has y – intercept 4 and is perpendicular to the line joining (2, -3) and (4, 2). **[3]**

(ii) The fourth term of the GP is square of its second term and the first term is -3. Determine the GP. **[3]**

(iii) A solid metallic sphere of radius 6cm is melted and made into solid cylinder of height 32cm. Find the: **[4]**

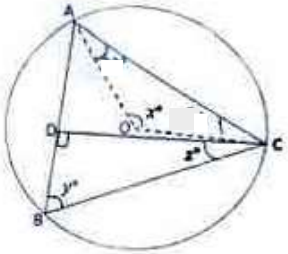
(a) radius of the cylinder.

(b) curved surface area of the cylinder.

[Take $\pi = 3.1$]

stion 9:

In the given figure, O is the center of the circle. CD is perpendicular to the chord AB, $\angle OAC = 32°$, find the value of x, y and z. **[3]**



(ii) A customer goes to the shop to buy an item quoted at ₹ 2000. The rate of GST is 12% on it. The customer asks for a discount on the price of that item to such an extent that she has to pay ₹ 2016 inclusive of GST.
Find the discount percentage on the price of that item. **[3]**

(iii) Draw a histogram of the following distribution and using it, calculate the mode. **[4]**

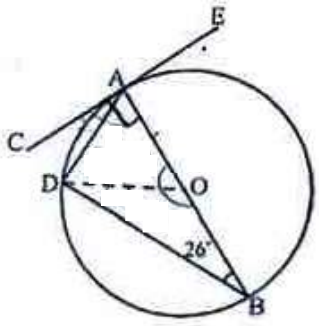
C I	0 - 10	10- 20	20 - 30	30 - 40	40 - 50	50 - 60
Frequency	5	15	10	5	12	8

Question 10:

(i) Prove that: $\frac{1}{1+\sin\theta} + \frac{1}{1-\sin\theta} = 2sec^2\theta$ **[3]**

(ii) Given: $\begin{bmatrix} 2 & 1 \\ -3 & 4 \end{bmatrix} X = \begin{bmatrix} 7 \\ 6 \end{bmatrix}$, Write **[3]**
(a) the order of the matrix X
(b) Find X.

(iii) In the given figure, O is the centre of the circle. CE is tangent to the circle at point A. If $\angle ABD = 26°$, then find: **[4]**
(a) $\angle BDA$
(b) $\angle BAD$
(c) $\angle CAD$
(d) $\angle ODB$



Question Paper 20

PRELIMINARY EXAMINATION

Grade : X Subject : MATHEMATICS Marks : 80

Time : 3 hours

- *You will **not be** allowed to write during the first **15 minutes**.*
- *This time is to be spent in reading the question paper.*
- *The time given at the head of this paper is the time allowed for writing the answers.*
- *This paper consists of 7 printed pages.*

- *Attempt all questions from **Section A** and **any four** questions from **Section B**.*
- *All working, including rough work, must be clearly shown and must be done on the same sheet as the rest of the answer.*
- ***Omission of essential working will result in loss of marks.***
- ***All figures in geometry have to be copied in the answer sheet.***
- *The intended marks for questions or parts of questions are given in brackets [ ].*

SECTION A

*(Attempt **all** questions from this Section)*

Question 1

Choose the correct alternative from the options given along with the questions: [15]

(i) If the 2nd term of a GP is $\frac{9}{4}$ and the 7th term is $\frac{8}{27}$, then the common ratio is

(a) $\frac{3}{2}$ (b) $\frac{2}{3}$

(c) $\frac{2}{7}$ (d) 1

(ii) The locus of the minute hand of a rectangular clock will be

(a) a rectangle (b) a circle

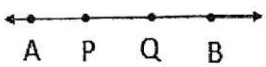
(c) an oval (d) a square

(iii) Meera invests in shares that are quoted at **₹ 150** each. The company declares a **dividend of 10%**. If Meera earns **8% return** on her investment, find the **face value** of each share

(a) 50 (b) 75

(c) 120 (d) 100

(iv) If P and Q are points of trisection of line segment joining A and B, then ratio of AP : AB is



(a) 3 : 1 (b) 1 : 2

(c) 2 : 1 (d) 1 : 3

(v) If $A=\begin{bmatrix}5 & 4\\ 3 & -1\end{bmatrix}$, $B=\begin{bmatrix}0 & 0\\ 0 & 1\end{bmatrix}$ and $C=\begin{bmatrix}2\\ 1\end{bmatrix}$

Which of the following statements are true ?

Statement I : (A+B) X C is not possible

Statement II: BA + C is possible

(a) only I (b) both I and II

(c) only II (d) neither I and II

(vi) If $d=\frac{x_i-45}{h}$, $\Sigma fd=28$, $\Sigma f=32$ and $\bar{x}=52$ then h (i.e class size) is

(a) 8 (b) 5

(c) 10 (d) 9

(vii) **Assertion (A)**: If 2k – 1 , 4k – 3 , 4k + 1 are three consecutive terms of an AP, then k = 1

Reason (R): If a, b, c are three consecutive terms of an AP then 2b = a + c

(a) Both assertion (A) and reason (R) are true but reason (R) is the correct explanation of assertion (A)

(b) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A)

(c) A is true, R is false

(d) A is false, R is true

(viii) In a showroom, a discount of 30% is offered on an article marked at ₹ 12,000. The rate of GST is 18%. The CGST to be paid is:

(a) ₹ 1512 (b) ₹ 1080

(c) ₹ 756 (d) ₹ 9156

(ix) The ratio of the side of a regular hexagon to the radius of its circumcircle is

(a) 6 : 1 (b) 1 : 2

(c) 1 : 1 (d) 1 : 6

(x) If the roots of the equations $ax^2+2bx+c=0$ and $bx^2-2\sqrt{ac}\,x+b=0$ are real, then which of the following conditions will satisfy both the above equations.

(a) $b=ac$ (b) $b^2=ac$

(c) $a^2=b^2c$ (d) $c^2=ab$

(xi) What should be added to $x^2+3x-30$ so that $(x-4)$ is a factor?

(a) -2 (b) 3

(c) -3 (d) 2

(xii) What additional information is required to make Δ ABC and Δ DEF similar if $\frac{AB}{DE}=\frac{BC}{FD}$.

(a) $\angle B=\angle D$ (b) $\angle A=\angle E$

(c) $\angle A=\angle F$ (d) $\angle B=\angle E$

(xiii) The length of a tangent from an external point P on a circle with centre O is

(a) always greater than OP (b) equal to OP

(c) always less than OP (d) equal to radius

(xiv) If $\sec\theta + \tan\theta = M$, then $\sec\theta - \tan\theta$ is equal to:

(a) $\frac{1}{M}$ (b) M

(c) $-M$ (d) 1

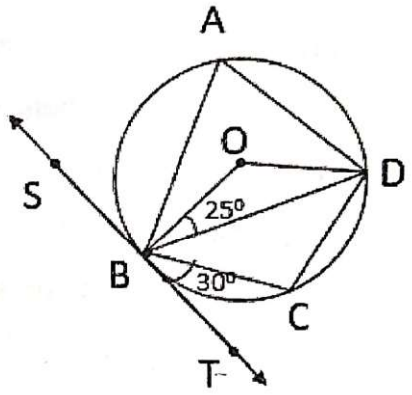
(xv) The volume of the largest cylinder that can be drilled out of a cube of side 2r is

(a) πr^3 (b) $2\pi r^3$

(c) $4\pi r^3$ (d) $3\pi r^3$

Question 2

(i) ABCD is a cyclic quadrilateral in the circle with centre O. ST is a tangent to the circle at point B. $\angle OBD = 25^0$ and $\angle CBT = 30^0$. Find : (a) $\angle BCD$ (b) $\angle ODC$. [4]

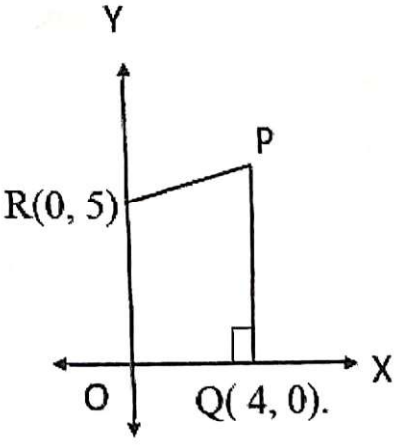


(ii) In the given diagram, PQ ⊥ x- axis. R(0, 5) and Q(4, 0). [4]

(a) Find the equation of line PQ.

(b) If slope of PR is 3, find the equation of line PR.

(c) Find the coordinates of P.



(iii) If $(x-2)$ is a factor of $2x^3 - x^2 - px - 2$ then, find the value of 'p'. [4]

Hence, factorise the given polynomial completely.

Question 3

(i) The product of first three terms of a GP is 512. When 2 is added to its second term, the three terms form an AP. Find the numbers. [4]

(ii) A metallic solid (figure I) is cut into two identical halves (figure II) along its height (as shown in the diagram). The diameter of the cylinder is 14 cm and the height is 30 cm. The height of the conical part is 24 cm. Find: [4]

(a) the total surface area (both the halves).

(b) the total cost of painting the two halves at the rate of 20 per cm^2 [5]

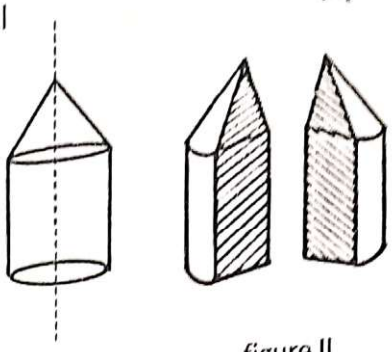


figure I figure II

(iii) Construct:

(a) an isosceles ΔABC in which AB = AC = 7 cm and BC = 6 cm.

(b) the locus of points such that it is 2.5 cm from the point A.

(c) the locus of points equidistant from B and C.

(d) Mark a point P closer to side BC which satisfies both the conditions mentioned in (b) and (c).

(e) a circle passing through P, B and C. Measure and record its radius.

SECTION B

*(Attempt **any four** questions from this Section)*

Question 4

(i) The terms 6, 13, 20, -------- 216 are in AP. Find [3]

(a) the number of terms in the given AP.

(b) the middle term

(c) the sum of first 16 terms

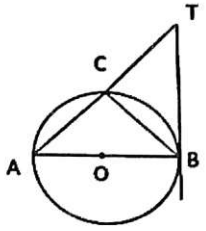
(ii) Mrs. Roy went to a departmental store and bought the following items : [3]

Items	Quantity	Rate (₹)	GST rate
Tata(salt)	2 kg	16/kg	0%
Tata Tea(tea)	500 g	250/kg	6%
Milk food(ghee)	1 Kg	350/kg	12%

She pays ₹ 1000 to the shopkeeper. How much money was returned to her?

(iii) AB is the diameter of a circle with centre O. [4]

The tangent at B and chord AC extended meet at T.



(a) Prove that Δ ABC ~ Δ ATB.

(b) If AB = 20 cm, AC = 16 cm, find TB.

(c) Find A(Δ ABC) : A(Δ ATB)

Question 5

(i) Arjav invested ₹24,000 in ₹ 50 shares at 4% discount giving 10% dividend. [4]
After one year he sold the shares at ₹65 and invested the proceeds (including his dividend) in ₹100 shares at ₹70 giving 11%. Calculate

(a) the change in annual dividend.

(b) the percentage increase in return on original investment.

(ii) A survey regarding height (in cm) of 60 boys belonging to class 10 of a school was conducted. The following data was recorded: [6]

Height (in cm)	135-140	140-145	145-150	150-155	155-160	160-165	165-170
No. of boys	4	8	20	14	7	6	1

Taking 2 cm = height of 10 cm along one axis and 2 cm = 10 boys along the other axis draw an ogive of the above distribution.

Use the graph to estimate the following:

(i) the median

(ii) lower quartile

(iii) If height above 158 cm is considered as the tall boy of the class, find the number of boys in the class who are tall.

Question 6

(i) Prove that $\sqrt{\frac{sec\,\theta-1}{sec\,\theta+1}} = cosec\,\theta - cot\,\theta.$ [3]

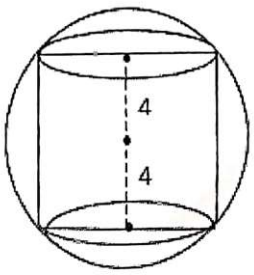
(ii) Using the properties of proportion, find the value of x.(x > 0) [3]

$$\frac{2x+\sqrt{3x^2+1}}{2x-\sqrt{3x^2+1}} = 15$$

(iii) Two hemispheres are joined together to just cover a cylinder of height 8 cm and base radius 3 cm. Find [4]

(a) the radius of the sphere

(b) the ratio of the volume of the remaining space to the volume of cylinder.



Question 7

(i) Shivangi deposits ₹ 500 every month in a recurring deposit scheme and receives ₹16,550 at the end of 2 ½ years. Calculate the rate of interest given by the bank. [3]

(ii) Solve the quadratic equation and write the answer correct to 2 decimal places : [3]

$(5x)^2 - 18x + 2 = 0$

(iii) Use a graph sheet for this question. Take 2 cm = 1 unit along both the axes. [4]

(a) Plot the following points A(0, –4) , B(3, –2), C(1, 0) , D(3, 2) & E(0, 1)

(b) Reflect the Points B , C , D on the line x = 0 to the points H, G, F respectively and write their co-ordinates.

(a) Join the points A, B, C , D , E , F , G , H and assign a geometrical name to the figure.

(d) Name two invariant points under line x = 0.

(e) Which point is the image of A under reflection on the line BH? Write its coordinates.

Question 8

(i) $A = \begin{bmatrix} 5 & -2 \\ 1 & 3 \end{bmatrix}$ *and* $B = \begin{bmatrix} 3 & -2 \\ 1 & 5 \end{bmatrix}$. If I is the identity matrix of 2 by 2 order. [3]

Evaluate: $A^2 + 4B - 5I$

(ii) In an equilateral triangle ABC , AD is the median. [3]

B = (6, 3) , C = (8, 5). Find the equation of AD.

(iii) A 112 cm wire is bent to form a right angled triangle with hypotenuse 50 cm. Find the length of the sides of the triangle. [4]

Question 9

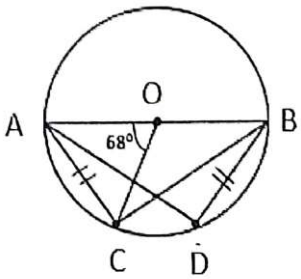
(i) O is the centre of the circle and AB is a diameter. [3]

If AC = BD and ∠AOC = 68^0,

Find (a) ∠ABC

(b) ∠BAD

(c) ∠ABD.



(ii) Solve the given inequation and graph its solution on the real number line. [3]

$4x - 19 < \frac{3x}{5} - 2 \leq \frac{-2}{5} + x,\ x \in N$